

School of Electronic Engineering
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Final Report

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Mathematics

Project Title:

**Linear Encoding and
Decoding of Brain
Activity in Response
to Music**

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Abstract

This undergraduate thesis investigates how the brain encodes and decodes acoustic features of music using linear temporal response function (TRF) modelling. Encoding models predict neural responses from the stimulus, while decoding models reconstruct the stimulus from neural responses. The analysis uses a publicly available electroencephalography (EEG) dataset of 20 participants (10 musicians and 10 non-musicians) listening to monophonic Bach pieces.

Six acoustic features were extracted from the stimuli: envelope, onsets, pitch, spectral centroid, mel spectrogram, and MFCC3. Two feature extraction pipelines, one using raw features and the other applying additional processing, were compared as an experimental variable. Encoding TRFs were fitted for all non-empty feature combinations per pipeline, and decoding TRFs for each feature individually, both across five EEG frequency bands. Performance was measured as the Pearson correlation between predicted and actual signals.

Envelope and onsets were decoded with the highest correlations, while spectral and cepstral features were reconstructed less reliably. Encoding correlations were lower than decoding correlations. The theta band gave the highest correlations across most features and both modelling directions. Musicians produced stronger neural responses and more right-lateralised decoder topographies than non-musicians across all five frequency bands. Adding features improved encoding performance once the mel spectrogram was excluded, and the processed pipeline outperformed the raw pipeline in both correlation and runtime.

Part of this work has been published at the 2025 Audio Engineering Society (AES) International Conference on Artificial Intelligence and Machine Learning for Audio (Bozilovic & Roman 2025), and presented as a talk and poster at the 2025 UCL NeuroAI Conference.

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Chapter 1: Introduction

1.1 Background

The auditory system is fundamental to how most humans interact with and interpret the world. From speech and **music** comprehension to environmental awareness, auditory processing enables the extraction of meaningful information from acoustic signals in real time (Bizley & Cohen 2013). A key challenge in auditory neuroscience is to understand how the brain encodes the temporal patterns present in sound (Wang et al. 2008).

Music is a particularly complex auditory stimulus, requiring the brain to engage in acoustic analysis, sensory memory, and the processing of emotion and meaning (Koelsch & Siebel 2005). Music can also reshape neural structures and improve well-being (Zatar et al. 2024) and most people experience music as inherently pleasurable (Vuust et al. 2022) without any conscious effort. Yet investigating music perception offers insights into how the brain works (Trimble & Hesdorffer 2017).

To study how the brain responds to music, the auditory signal is typically first represented in a form that can be more easily related to neural activity (Crosse et al. 2021). One common way to do this is to extract **acoustic features**, quantitative descriptors that capture different aspects of the signal, such as its overall amplitude (envelope), the timing of new events (onsets), or perceived highness of a sound (pitch) (Alías et al. 2016). These features serve as input to computational models that link the stimulus to the recorded brain response (Crosse et al. 2016, Di Liberto et al. 2020).

On the brain-response side, **electroencephalography (EEG)** is a non-invasive method for recording electrical activity from the scalp with high temporal resolution (Teplan et al. 2002). The recorded EEG signal can be divided into **frequency bands**: delta (1–4 Hz), theta (4–8 Hz), alpha (8–12 Hz), and beta (12–30 Hz). Different bands have been linked to different aspects of auditory processing, which affects how clearly the neural response to a continuous stimulus can be recovered (Giraud & Poeppel 2012, Galeano-Otálvaro et al. 2024).

Numerous studies have examined how acoustic features are transformed into brain activity using EEG recordings (Galeano-Otálvaro et al. 2024, Lalor & Foxe 2010, Di Liberto et al. 2020). **Encoding (forward) models** are trained to predict neural signals from audio, while **decoding (backward) models** attempt to reconstruct a stimulus from recorded neural activity (Brodbeck et al. 2023).

These approaches deepen our understanding of how audio becomes brain activity, and also open possibilities for identifying disruptions in this process, such as in clinical conditions that affect auditory cognition (Crosse et al. 2021). Beyond research, neural decoding has begun to inform assistive technologies and neurological rehabilitation (Moses et al. 2021). Furthermore, new directions in neural audio engineering for brain-computer interfaces are being explored, where imagined sound can be reconstructed into audio (Lee et al. 2023, Di Liberto et al. 2021).

1.2 Motivation and aim

The motivation for this research is to understand how auditory stimuli are transformed into neural signals, and to what extent this transformation can be modelled, predicted, and inverted. The **Temporal Response Function (TRF)** is a linear model that maps the relationship between continuous stimuli and neural responses (Crosse et al. 2016). This thesis applies TRFs to perform both forward and backward modelling of neural responses to music. The aim is to investigate the effect of different acoustic features through single-feature decoding and through single and multi-feature encoding, including combinations that have not been examined in previous music–eeg TRF studies. A further aim is to identify the best-performing frequency bands of the EEG data and to determine whether there is a difference between musicians’ and non-musicians’ responses to music across these models.

1.3 Research questions and hypotheses

1.3.1 Research questions

This project answers the following research questions:

- RQ1: Can the melody someone is listening to be reconstructed from their neural data?
- RQ2: Can neural data be predicted from the music someone is listening to?
- RQ3: Does increasing the number of acoustic features in encoding models improve prediction performance?
- RQ4: How does encoding and decoding performance vary across frequency bands of the neural data?
- RQ5: Does musical expertise lead to measurable differences in the neural responses of music?

1.3.2 Hypotheses

The corresponding hypotheses are:

- The melody someone is listening to can be reconstructed from their neural data to a meaningful extent.
- Neural data can be predicted from music to a meaningful extent.
- Adding acoustic features to encoding models improves prediction performance.
- Encoding and decoding performance is the highest in the low frequency bands.
- Musicians show enhanced neural responses to music compared with non-musicians.

1.4 Problem Statement

TRF modelling has been used widely to study neural responses to audio, but most existing work either applies forward modelling alone or only focuses on speech data (Te Rietmolen et al. 2024, Kern et al. 2022, Mesik & Wojtczak 2023). Comparisons of

how different acoustic features and their combinations contribute to TRF performance across frequency bands are limited (Hołubowska et al. 2026, Bianco et al. 2026), and the choice of feature extraction pipeline is rarely treated as an experimental variable. As a result, it remains unclear which acoustic representations best capture the stimulus–brain relationship in music, how performance varies across frequency bands, and how these factors interact with musical expertise.

This project addresses these gaps by performing encoding and decoding TRF analysis on music EEG data, comparing single and combined acoustic features (for encoding), multiple frequency bands, two feature extraction pipelines, and musician versus non-musician groups.

1.5 Preliminary Requirements

1.5.1 Data

This project uses the publicly available dataset from Di Liberto et al. (2020), which provides:

- EEG recordings of 20 participants listening to musical stimuli, with both musicians and non-musicians represented
- MIDI files of the stimuli, 10 monophonic Bach pieces (used to render audio from which acoustic features are extracted)

1.5.2 Software

The following software tools are required:

- **Python** – primary programming language
 - **NumPy** and **SciPy** – numerical computation
 - **PrettyMIDI** and **Soundfile** – rendering of MIDI files into audio (WAV files)
 - **Librosa** – acoustic feature extraction
 - **MNE-Python** – EEG loading and preprocessing
 - **Eelbrain** – TRF modelling, cross-validation and prediction evaluation
 - **Pandas** – aggregation and management of results
 - **Statsmodels** and **Pingouin** – statistical analysis
 - **Matplotlib** – figure generation

1.5.3 Hardware

The TRF estimation pipeline is computationally expensive: each subject requires fitting hundreds of forward and backward TRF models (see Chapter 3 for details). `Eelbrain`'s algorithm used is implemented for multi-core CPU execution rather than GPU acceleration (Brodbeck et al. 2023), so performance scales with the number of available CPU cores and the amount of available RAM. Together, these factors made the pipeline impractical to run on a standard local machine within the project's timeframe.

It was therefore run on Amazon Web Services (AWS) compute instances, which provided the high CPU and RAM capacity required to process all 20 subjects in parallel in a reasonable time. The pipeline can also be executed locally, but with considerably longer runtimes.

1.6 Objectives

The following objectives must be accomplished in order to achieve the aim of this project:

- To review the literature on neural responses to music and TRF modelling
- To prepare the data:
 - Render audio from MIDI files and extract acoustic features
 - Preprocess the EEG and split it into five frequency bands
- To implement forward (encoding) and backward (decoding) TRF pipelines
- To evaluate single-feature decoding and single and multi-feature encoding across frequency bands
- To compare TRF performance between musicians and non-musicians

1.7 Report Structure

The remaining chapters are organised as follows. Chapter 2 reviews the relevant literature on neural responses to music and TRF modelling. Chapter 3 describes the methodology, including the dataset, feature extraction, EEG preprocessing and TRF modelling. Chapter 4 presents the results across acoustic features, frequency bands, musical expertise groups and pipelines. Chapter 5 discusses the findings and analyses the strengths and limitations. Chapter 6 concludes and suggests directions for future work.

Chapter 2: Literature Review

2.1 Stimulus Representation: Acoustic Features

2.1.1 Sound and Music

Sound is vibrational energy that travels as waves through a medium, in this case, air (Taylor 2000). When these vibrations reach the ear, they pass through the outer and middle ear to the cochlea, where they are converted into neural signals. The frequency of these vibrations is measured in Hertz (Hz), where one Hertz equals one vibration per second. Most sounds contain energy at many frequencies at once and the distribution of this energy across frequency is called the sound's spectrum. Humans can typically hear frequencies between 20Hz and 20kHz (Hunter et al. 2020). Within this range, sounds fall into several broad categories that the auditory system processes with different priorities.

Environmental sounds, such as footsteps, animal calls, or alarms, span a wide range of frequencies and are recognised mainly through their amplitude envelope and onsets (Shafiro 2008). They help listeners stay aware of their surroundings.

Speech carries linguistic information through phonemes, syllables and phrases, which unfold at different rates and contain most of their energy at low frequencies (Giraud & Poeppel 2012). It allows most humans to communicate with one another.

Music consists of pitched tones organised into melodies (sequences of notes), harmonies (notes played together), and rhythms (patterns of timing) (Koelsch & Siebel 2005, Vuust et al. 2022). Different instruments and voices also have distinct timbres (tone 'colours') which is why a piano and a flute sound different even when playing the same note. Music is found across cultures, and listening to it is often a source of pleasure and emotional experience (Vuust et al. 2022). Of these three categories, this thesis focuses on music.

2.1.2 Features of Music

To analyse music computationally, it is first stored digitally as waveforms. However, waveform itself is too detailed to use directly when modelling brain responses, since the brain follows only the slower changes in a sound rather than every fluctuation in the signal. Researchers therefore extract different features, which capture specific properties of the sound over time (Crosse et al. 2021).

These features are separated into two broad categories. Acoustic features describe physical properties of the sound, such as envelope, onsets, and pitch. Higher-level features capture musical structure rather than the raw acoustics. One example is melodic expectation, a measure of how predictable each note is given the previous ones, derived from statistical models of music such as IDyOM (Information Dynamics of Music) (Pearce & Wiggins 2012, Di Liberto et al. 2020). This thesis focuses on acoustic features.

Acoustic features can be grouped further by what aspect of the signal they capture. Temporal features describe how energy of the signal varies over time, spectral features describe how energy is distributed across frequency, and cepstral features describe

the shape of the spectral envelope, the overall contour of the signal's spectrum. (Alías et al. 2016, Peeters et al. 2011).

This thesis focuses on six acoustic features, described below: envelope and onsets (temporal); pitch, spectral centroid, and mel spectrogram (spectral); and MFCC3 (cepstral). Although these features can be extracted from any sound, here they are applied to musical stimuli.

2.1.3 Temporal Features

Amplitude envelope tracks slow fluctuations in sound energy over time, and is the most widely studied stimulus feature in EEG research. It was shown that the envelope is reliably tracked by neural responses recorded in audio-EEG studies (Lalor & Foxe 2010), and it remains a benchmark feature against which others are compared (Crosse et al. 2016, 2021).

Onsets mark the points in time at which new sound begins. They are particularly informative in music, since note onsets carry the rhythmic and temporal structure of the music signal (Di Liberto et al. 2020). Onsets are typically derived from the envelope by finding sharp positive increases in the amplitude. The envelope and onsets capture complementary aspects of the signal and thus they are commonly used together in TRF models (Brodbeck et al. 2023, Drennan & Lalor 2019).

2.1.4 Spectral and Cepstral Features

Pitch corresponds to how high or low a sound seems to a listener. It is set by the lowest frequency in the sound signal, the fundamental frequency (F0), which determines the note that is heard. The sequence of pitches over time forms a melody (Sankaran et al. 2024).

Spectral centroid is the weighted average frequency of a sound's spectrum, showing if its energy sits more towards lower or higher frequencies. It corresponds to the 'brightness' of a sound (for example a flute sounds brighter than a tuba), which is one dimension of timbre (Allen et al. 2019).

Mel spectrogram captures how the energy at different frequencies in a sound changes over time, with frequency bands spaced on the mel scale (Haupt et al. 2025). The mel scale stretches out low frequencies and compresses high frequencies, matching how humans perceive pitch differences more clearly at low than at high frequencies (Guest & Oxenham 2022).

Mel-frequency cepstral coefficients (MFCCs) are derived from the mel spectrogram using a mathematical function (discrete cosine transform, DCT) that compresses it into a set of coefficients summarising the shape of the spectrum (Alías et al. 2016). They are commonly used as a descriptor of timbre. Lower-order coefficients describe the broadest features of the spectral envelope, while higher-order coefficients capture finer detail. MFCC0 captures the average energy across frequency bands, MFCC1 captures the spectral tilt (whether energy leans more towards low or high frequencies), and MFCC2 captures spectral curvature (whether energy is concentrated in the middle of the spectrum or at the extremes). **MFCC3** captures a three-region alternating pattern across the spectrum (low, medium, and high), reflecting a more detailed

spectral shape than the tilt and curvature represented by MFCC1 and MFCC2. This three-region pattern comes from each MFCC corresponding to a cosine wave with a different number of half-cycles across the spectrum. MFCC3 corresponds to a cosine with three half-cycles (and thus three regions). The same principle applies to each order MFCCs: each one adds one more half-cycle than the previous coefficient.

In this thesis, MFCC3 is used as a single time-varying feature. Lower-order coefficients were not included as separate features because they largely overlap with information already captured by other features in this thesis: MFCC0 describes overall energy, which overlaps with envelope, while MFCC1 and MFCC2 describe spectral tilt and curvature, which overlap with information captured by the centroid. Higher-order coefficients (MFCC4 and above) describe finer spectral detail that is more informative for complex or harmonically rich timbres. Since the stimuli in the dataset used are monophonic pieces and synthesised with a single piano timbre, this finer detail varies little over time and is unlikely to add meaningful information for TRF modelling.

Including such spectral and cepstral features alongside envelope and onsets allows TRF models to capture aspects of music that temporal descriptors only cannot.

2.2 Neural Data: Electroencephalography (EEG)

2.3 Neurons, brain imaging and EEG

Neurons communicate with each other at synapses, where neurotransmitters (chemical signals) released by one cell bond to receptors on a neighbouring cell and cause small electrical changes, postsynaptic potentials, in that cell. When many neurons in the same region produce these potentials at the same time, their combined activity generates electrical fields that can be measured non-invasively (Teplan et al. 2002).

Several brain imaging techniques have been developed to record this kind of brain activity:

- Functional magnetic resonance imaging (fMRI) measures changes in blood oxygenation linked to neural activity (Glover 2011).
- Magnetoencephalography (MEG) measures the small magnetic fields generated by postsynaptic potentials in neurons (Hämäläinen et al. 1993).
- Electroencephalography (EEG) measures electrical activity directly from the scalp using surface electrodes. The recorded signal reflects the summed postsynaptic potentials of large populations of neurons (Teplan et al. 2002).

Its high temporal resolution makes EEG a suitable tool in research on neural responses to continuous stimuli such as music (Galeano-Otálvaro et al. 2024). This thesis analyses the Di Liberto et al. (2020) EEG dataset.

2.3.1 Electrode Placement

EEG is recorded using multiple electrodes arranged across the scalp according to standardised placement systems. The original 10-20 system positions electrodes at intervals of 10% and 20% of the distance between skull landmarks (the nasion at the bridge of the nose, the inion at the back of the skull, and the two preauricular points

in front of the ears) (Seeck et al. 2017). High-density recordings typically use the **10-10 system**, which adds intermediate electrode positions at 10% intervals throughout, replacing the 20% gaps of the original system (Oostenveld & Praamstra 2001).

Since EEG measures voltage differences, a **reference electrode** is required. Common choices include the average of mastoids (electrodes placed on bony areas behind the ears) or the average of all electrodes (Yao et al. 2019).

2.3.2 Frequency Bands

EEG activity can be divided into frequency bands. Following Galeano-Otálvaro et al. (2024), this thesis examines four frequency bands: delta (1–4 Hz), theta (4–8 Hz), alpha (8–12 Hz), and beta (12–30 Hz). Additionally, the thesis examines wideband (1–30 Hz) which combines all four bands into a single one.

Different EEG frequency bands have been associated with distinct functions in music processing (Giraud & Poeppel 2012):

- **Delta** activity follows the slow rhythmic structure of music, such as the beat (Nozaradan et al. 2012). It has been linked to the processing of note onset predictions, particularly when onsets will occur (Galeano-Otálvaro et al. 2024).
- **Theta** activity follows faster changes in music, such as the rate of individual notes. Its tracking of music has been shown to be stronger in musicians than in non-musicians (Doelling & Poeppel 2015).
- **Alpha** activity does not directly follow the music. Instead, it decreases when the music is more predictable or when the listener is more engaged with it (Klimesch 1999, Malekmohammadi et al. 2023). It has been linked to the processing of musical expectations in non-musicians (Galeano-Otálvaro et al. 2024).
- **Beta** activity does not follow the music either. Instead, it rises before expected notes and links the auditory and motor systems, which has been linked to predictive timing and moving in synchrony with a rhythm (Fujioka et al. 2012, Galeano-Otálvaro et al. 2024).

2.3.3 Preprocessing

Proper preprocessing is essential for reliable analysis of stimulus-related neural activity. Preprocessing pipelines for EEG typically include (Crosse et al. 2021):

- Bandpass filtering – to keep the frequencies of interest (for example 1–30 Hz for auditory studies)
- Downsampling – to reduce computational cost during model training (for example from 500 Hz to 100 Hz)
- Re-referencing – to enhance neural responses to stimuli of interest
- Removal or interpolation of channels with excessive noise – to prevent bad channels from affecting the rest of the analysis
- Artifact removing – to separate neural activity from noise such as activity from

blinks, heartbeat and facial muscles (Jiang et al. 2019, Croft & Barry 2000, Goncharova et al. 2003, Tamburro et al. 2019)

2.4 Temporal Response Functions (TRFs)

2.4.1 Approaches to Audio–EEG Modelling

A key methodological choice in this thesis was to use linear TRFs instead of non-linear models such as artificial neural networks. The literature on this trade-off is mixed. Ciccarelli et al. (2019) found that a non-linear decoder did not outperform a linear one on stimulus reconstruction. The improvements in the study came from changing the task to classification. Accou et al. (2023) reported a 52% improvement over a linear decoder using their VLAAI network, but accuracy remained modest (median correlation coefficient $r = 0.19$). Puffay et al. (2023) reviewed 29 deep-learning studies relating EEG to continuous speech and found that non-linear improvements are not yet a general finding, with many studies showing inflated performance from biased cross-validation and overfitting.

Linear models can be interpreted directly as the brain’s response to each feature, showing when (at which time lags) and where (at which scalp locations) the brain reacts. Non-linear models are harder to interpret. They also typically require more data and computation time, and the literature does not yet show that such an investment would reliably improve results (Puffay et al. 2023).

Thus, this thesis focuses on linear modelling, and uses the TRF as the most widely used linear method in audio–EEG research (Crosse et al. 2021, Di Liberto et al. 2020, Galeano-Otálvaro et al. 2024).

2.4.2 The Encoding Model

The TRF analysis assumes that the neural response at any time point can be modelled as a linear convolution of the stimulus with a filter (the TRF) plus noise:

$$r(t) = \sum_{\tau} s(t - \tau) w(\tau) + \epsilon(t) \quad (1)$$

where $r(t)$ is the neural response (EEG) at time t , $s(t)$ is the stimulus feature (for example envelope), $w(\tau)$ is the TRF which weights the contribution of the stimulus at time lag τ (the delay between the stimulus and its corresponding neural response), and $\epsilon(t)$ is the residual error (Crosse et al. 2016).

In the forward (encoding) model, the TRF maps stimulus features to neural responses. This allows researchers to determine which stimulus features are encoded in neural activity and at what latencies. When more than one stimulus feature is included as input, the model is referred to as a **multivariate TRF (mTRF)**, and it learns separate TRF weights for each feature simultaneously (Crosse et al. 2016).

2.4.3 TRF Weights Estimation

The TRF weights are estimated from the data (EEG and acoustic features). A widely used method is ridge regression which solves for all TRF weights at once and adds a

penalty on large weights to prevent overfitting. The optimal value of the regularisation parameter (the strength of the penalty) is usually chosen via cross-validation (Crosse et al. 2016, 2021).

An alternative is the boosting algorithm, an iterative method that starts with all TRF weights set to zero, updates them one at a time, and stops when the validation error stops decreasing. Since boosting leaves most weights at zero, redundant predictors are left out which makes it well suited for mTRFs with correlated predictors (Brodbeck et al. 2023). This is why it was chosen as the estimation method for this thesis. The algorithm is described in detail in Chapter 3.

2.4.4 The Decoding Model

In the backward (decoding) model, the direction is reversed: the model reconstructs stimulus features from neural activity. This is achieved by learning a decoder that maps from EEG to the stimulus:

$$\hat{s}(t) = \sum_{\tau} r(t + \tau) \cdot d(\tau) \quad (2)$$

where $\hat{s}(t)$ is the reconstructed stimulus and $d(\tau)$ is the decoder (Crosse et al. 2016).

The decoder $d(\tau)$ weights are estimated in a similar way to the encoder, using either ridge regression or boosting (Crosse et al. 2016, Brodbeck et al. 2023). Unlike the encoding model, which explains the measured (true) EEG as the convolution of the stimulus with the TRF plus noise, the decoding model produces an estimate (reconstruction) of the stimulus, so no noise term is included.

Decoding models are multivariate by construction, since the decoder takes all EEG channels as input simultaneously and learns how to combine them to reconstruct the stimulus (Crosse et al. 2016).

2.4.5 Music–EEG TRF Studies

TRF analysis has been applied to both speech and music EEG data, but most existing studies focus on a narrow set of features (and their combinations), frequency bands, or modelling directions.

TRF EEG research has been more widely studied with speech (Mesik & Wojtczak 2023, Di Liberto et al. 2015, Lalor & Foxe 2010), where envelope has been studied extensively, often in one direction only (usually encoding).

Di Liberto et al. (2020) introduced the dataset analysed in this thesis. Using TRF encoding with ridge regression, they showed that EEG responses to music could be predicted from envelope and onsets, with the addition of melodic expectation features (derived from the IDyOM) significantly improving prediction accuracy. However, this comparison only contrasted two feature sets (acoustic features alone, and acoustic features combined with melodic expectations), without exploring how different combinations of acoustic features contribute to encoding performance. Also, although they noted that broader bands did not change their overall results, they restricted their main

analysis to the combined 1–8 Hz low-rate band, without separately examining performance in delta and theta. This leaves open whether these individual bands carry distinct music-specific information. Finally, they performed forward modelling only, which means it remains unknown to what extent the features can be reconstructed from the recorded EEG.

In a study closely related to this thesis, Galeano-Otálvaro et al. (2024) re-analysed the EEG dataset of Di Liberto et al. (2020) using mTRF and showed that melodic expectations enhanced EEG reconstruction accuracy across all frequency bands below the gamma range (below 30 Hz). However, by restricting the encoding model to envelope, onsets, and melodic expectation features (and combinations of these), this work did not show how other acoustic features of music (pitch, centroid, etc.) contribute to the neural response.

The dataset (Di Liberto et al. 2020) has been used in further encoding studies (Galeano-Otálvaro et al. 2024, Kern et al. 2022), but decoding from this data has not been investigated. Existing music decoding work using TRF tends to focus on a single feature (typically envelope) (Di Liberto et al. 2021), leaving open whether some features are more reliably represented in EEG than others.

The effect of musical expertise on TRF performance has also been examined. Di Liberto et al. (2020) reported larger encoding of melodic expectations in musicians than in non-musicians at frontal electrodes. Galeano-Otálvaro et al. (2024) reported that melodic expectations enhanced EEG reconstruction in the beta band for musicians but in the alpha band for non-musicians. However, these comparisons have been limited to envelope, onsets, and melodic expectations, leaving open whether expertise effects extend to other acoustic features and to decoding models.

This thesis addresses gaps mentioned. Using the Di Liberto et al. (2020) dataset, it performs both forward (encoding) and backward (decoding) TRF analysis across five frequency bands (delta, theta, alpha, beta, and the wideband), six acoustic features (envelope, onsets, pitch, centroid, mel spectrogram and MFCC3), and a range of multi-feature combinations not previously examined for encoding. It also compares two feature extraction pipelines, treating the pipeline itself as an experimental variable, and compares musicians with non-musicians across all of these conditions.

Chapter 3: Methodology

3.1 Dataset

The dataset (Di Liberto et al. 2020) comes from an experiment in which 20 participants passively listened to 10 melodies while their EEG was recorded. This thesis uses the most recent version of the dataset, uploaded to the Dryad repository on 8 October 2021.

Participants were separated into two groups: musicians, who were 10 expert pianists with a degree in music and over 10 years of experience, and non-musicians, who had no musical background. All participants were healthy individuals and out of the 20 subjects, 10 were male and 10 female, aged between 23 and 42 with a mean age of 29 (Di Liberto et al. 2020).

The melodies were 10 monophonic Johann Sebastian Bach pieces played on a piano (average duration: 2min 40s per piece). The dataset provides the 10 MIDI files corresponding to these stimuli, along with the resulting EEG recordings for each participant. Each piece was presented three times, thus in total each participant listened to 30 musical pieces, corresponding to 30 trials per participant.

The EEG was recorded at 512 Hz using a BioSemi ActiveTwo system (a high-density EEG amplifier), with 64 scalp electrodes arranged according to the 10–10 system. Two mastoid electrodes were also recorded and used as the reference (the EEG was re-referenced to the average of the two mastoids) (Di Liberto et al. 2020).

3.2 Stimulus Preparation

3.2.1 Audio Rendering

10 audio files in Waveform Audio File Format (WAV) were rendered from the 10 MIDI stimuli using PrettyMIDI's built-in synthesiser at a sampling rate of 44,100 Hz, matching the rate at which the original stimuli were presented in the EEG experiment (Di Liberto et al. 2020). Since PrettyMIDI's synthesiser can produce a small amount of leading silence before the first audible note, this silence was trimmed using a small amplitude threshold of 10^{-4} . Each waveform then began at the first sample whose absolute amplitude exceeded this value. This step ensured that the rendered audio aligned with the start of the corresponding EEG trial. Each waveform was then amplitude-normalised, ensuring that all 10 stimuli had a consistent loudness range and that subsequent feature extraction was not biased by differences in absolute amplitude across musical pieces. These WAV files were then used as input for acoustic feature extraction.

3.2.2 Acoustic Feature Extraction

Six acoustic features were extracted from the WAV files: envelope, onsets, pitch, spectral centroid, mel spectrogram, and MFCC3. All features were computed at a target sampling rate of 100 Hz, achieved by setting the hop length (the number of audio samples between successive frames of the extracted feature) to 441 samples at a 44,100 Hz audio sampling rate, giving exactly 100 frames per second.

Two feature extraction pipelines were implemented. The **raw pipeline** extracted features as returned by the Python libraries with no processing after. The **processed pipeline** applied additional transformations to several features (described below) and also z-scored each feature across all 10 stimuli combined (subtracting the mean and dividing by the standard deviation, computed across all 10 stimuli).

The raw pipeline produced the following features:

- **Envelope** is computed as the Hilbert envelope of the WAV signal (using the Hilbert transform) in *Eelbrain*.
- **Onsets** are derived from the envelope by taking the first time derivative and keeping only the positive part, so that sharp positive increases in amplitude appear as peaks while decreases are set to zero.
- **Pitch** is computed as the fundamental frequency (F0) using the pYIN algorithm (Mauch & Dixon 2014), a probabilistic algorithm that tracks the most likely F0 trajectory over time, in *Librosa*. For frames where no pitched sound is detected, pYIN returns NaN values and these were filled by linear interpolation. The frequency bounds for pitch were set to E3–C7 (a three-semitone below lowest and above highest notes in the all stimuli).
- **Spectral centroid** is computed as the weighted average (mean) frequency of the sound's spectrum at each frame in *Librosa*.
- **Mel spectrogram** is computed in *Librosa* by splitting the audio into 2048 sample frames (short pieces of the audio used for analysis) and applying the Fast Fourier Transform (FFT, which converts each frame into its frequency components) to each frame. The audio is split into frames first because the FFT gives the frequency content of an entire signal at once, but using short frames instead means it can capture how frequency changes over time. The resulting power spectrum is then mapped onto 8 mel-scale frequency bands.

The processed pipeline started from the same envelope, onsets, pitch, and centroid as the raw pipeline, with the following modifications applied before z-scoring:

- **Envelope** is unchanged (apart from the final z-scoring).
- **Onsets** are computed as in the raw pipeline, then smoothed with a 10 Hz low-pass Butterworth filter, which keeps slow changes and removes the fast ones (small spikes visible in the 'Onsets (raw)' in Figure 1 below), leaving only prominent peaks at note onsets. Any negative values introduced by the filtering are clipped to zero.
- **Pitch** is computed as in the raw pipeline, then represented as the absolute first time derivative of F0 ($|dF_0/dt|$), giving a spike at each note change rather than the continuous F0 trajectory (as shown by 'Pitch (processed)' in Figure 1 below).
- **Spectral centroid** is unchanged (apart from the final z-scoring).
- **MFCC3** is computed in *Librosa* by applying the DCT to the mel spectrum (from the same 2048 sample frames as the mel spectrogram) and keeping the MFCC3 coefficient. MFCC3 was used instead of the mel spectrogram as it captures three-region spectral pattern in a single time-varying coefficient.

Figure 1 shows an example of these features.

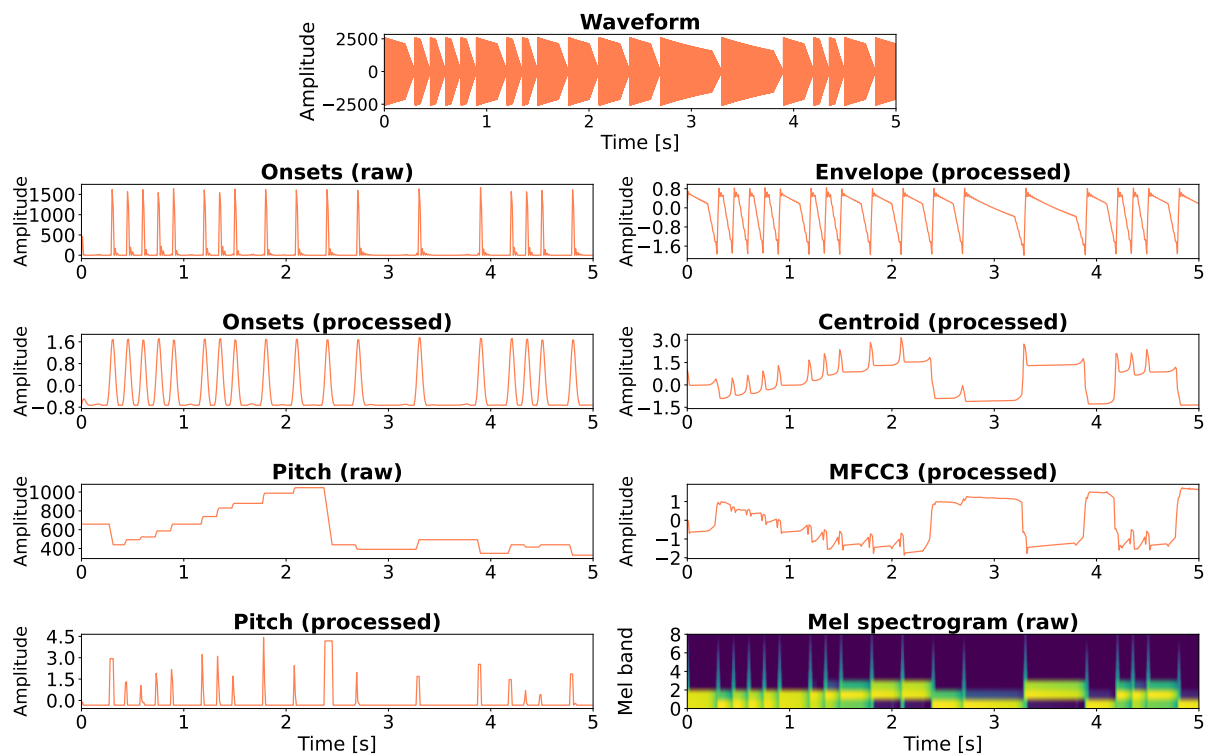


Figure 1: Acoustic features and waveform for the first 5 seconds of the second musical piece. Top: audio waveform. Remaining panels: the six acoustic features used in this thesis. Onsets and pitch are shown in both raw and processed forms to illustrate the effect of the processing.

3.3 EEG Preprocessing

EEG preprocessing was implemented using `MNE-Python` and applied identically across all subjects, frequency bands, and feature pipelines. The dataset provides the EEG segmented into 30 trials per participant. The 64 scalp channels of each trial were extracted and rescaled into microvolts (the standard EEG measurement unit). Channel labels and scalp positions of the channels were also loaded from the dataset to enable topographic visualisation later.

The 30 trials were then concatenated into a continuous stream (also in `MNE-Python`), with a stimulus channel added marking the onset of each trial so that the trials can be re-segmented later. Concatenating the trials into a continuous stream before filtering avoids edge artifacts that bandpass filtering would otherwise produce at each trial's start and end (Crosse et al. 2021).

The continuous EEG was then bandpass-filtered into one of five frequency bands using a default zero-phase FIR (Finite Impulse Response) filter which preserves the timing alignment between the EEG and the stimulus (by applying the filter both forwards and backwards which cancels out the time delay):

- **Delta:** 1–4 Hz
- **Theta:** 4–8 Hz

- **Alpha:** 8–12 Hz
- **Beta:** 12–30 Hz
- **Wideband:** 1–30 Hz

After filtering, the trial onset markers were extracted as `Eelbrain` events, and each event was then paired with the acoustic features extracted from the corresponding musical piece. The EEG was then re-segmented into the 30 trials. The trial onset markers identified where each trial began in the continuous stream, while the corresponding envelope's duration determined how long each trial lasted (since the 10 musical pieces had different durations). During this step, the EEG was downsampled to 100 Hz to match the sampling rate of the acoustic features.

Figure 2 shows an example of preprocessed EEG.

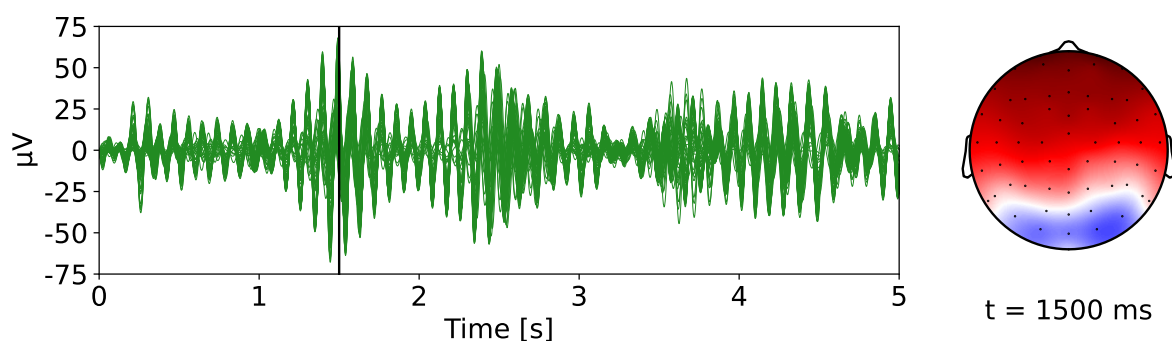


Figure 2: First 5 seconds of preprocessed EEG (alpha band) from the first trial of the second piece for Subject 19. Left: butterfly plot showing all 64 channels (zoom in for individual channels). Right: topographic map of the voltage distribution across the scalp at $t = 1500$ ms.

3.4 TRF Modelling

Several steps in the implementation (EEG bandpass filtering, pairing events with stimulus features, and per-trial normalisation (explained below)) followed a reference TRF pipeline provided in the `Eelbrain`'s documentation for the 'Alice EEG dataset' (these lines are marked in the code with the comment "TRF for Alice EEG Dataset") (Brodebeck et al. 2023).

3.4.1 Boosting Function

The TRF weights were estimated using the boosting algorithm implemented in `Eelbrain`'s `boosting()` function for both encoding and decoding. As described in the Chapter 2, boosting starts with all TRF weights set to zero and iteratively updates them one at a time. At each iteration, the algorithm searches across all time lags (and predictors, in the multivariate case) to find the single weight whose update by a fixed increment δ would most reduce the prediction error (the difference between the model's prediction and the actual data) on the training data. After each update, performance is checked on a held-out validation set (a subset of the data not seen during training), and iterations stop when the validation error stops decreasing. This early

stopping prevents overfitting and tends to leave most weights at zero, since only the most informative ones get updated.

The procedure behind the boosting function is summarised in pseudocode below:

```

initialise weights[predictor, lag] = 0 for all (predictor, lag)
while validation_error is decreasing:
    for each (predictor, lag):
        delta_error[predictor, lag] = change in training_error
            from updating weights[predictor, lag] by +/-delta
    (best_predictor, best_lag) = argmin(delta_error)
    weights[best_predictor, best_lag] += +/-delta
    validation_error = evaluate(weights) on validation_set
return weights

```

3.4.2 TRF Hyperparameters

The same TRF hyperparameters were used for both the raw and processed feature pipelines, varying only in the input features used. The hyperparameters were:

- **Cross-validation:** 4 partitions with held-out test set (`partitions=4, test=True`). At each fold, one partition is used as the test set, one as the validation set (for early stopping), and the remaining two for training. This is repeated 4 times so that every partition serves as the test set once, producing cross-validated test predictions for all 30 trials.
- **Error:** L1 (`error='l1'`). The L1 error is the sum of absolute differences between the model's prediction and the actual data. It is used to evaluate which weight update reduces the error most.

Basis function: 50 ms Hamming window basis (`basis=0.050`). At the 100 Hz sampling rate, the TRF has one weight every 10 ms. Instead of estimating each of these weights independently, boosting builds the TRF from overlapping 50 ms Hamming windows, which smooths the estimated weights and reduces sensitivity to noise at individual time lags.

- **Encoding time lags:** -150 to $+750$ ms (`tstart=-0.150, tstop=0.750`). Since the stimulus precedes the neural response, the TRF mostly uses positive lags (post-stimulus), with a smaller negative range to capture any pre-stimulus activity.
- **Decoding time lags:** -600 to $+200$ ms (`tstart=-0.600, tstop=0.200`). Since the EEG response follows the stimulus, the decoder relies on past EEG (negative lags) to reconstruct it, with a smaller positive range to capture any remaining relationship.

3.4.3 Encoding Models

For each subject, frequency band, feature combination, and pipeline a separate encoding TRF model was trained to predict the EEG signal from acoustic features. All 31 non-empty subsets of the five features were used as inputs for each pipeline, giving 31 encoding models per subject per band per pipeline.

Per-trial prediction performance was computed by convolving the trained encoder with the trial's normalised features to predict the EEG, then computing the Pearson correlation (r) between predicted and actual EEG, averaged across the 64 channels.

Figure 3 shows an example encoding TRF.

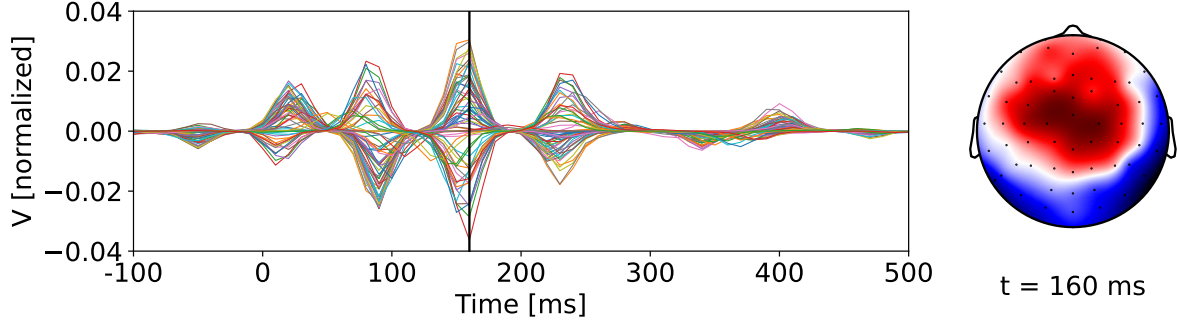


Figure 3: Encoding TRF weights for Subject 13, processed envelope in theta band. Left: butterfly plot showing the TRF across all 64 channels (flat lines are channels left at zero by boosting) from -100 to $500ms$, where channels show meaningful activity. Right: topographic map across the scalp at the latency where the TRF response is largest across channels, with red indicating positive and blue indicating negative voltage.

3.4.4 Decoding Models

For each subject, frequency band, feature, and pipeline a separate decoding TRF model was trained to reconstruct individual acoustic features from the EEG. Unlike encoding, decoding was performed one feature at a time, since Eelbrain's boosting decodes each output independently regardless of whether features are passed jointly or separately. Thus, there are 5 decoding models per subject per band per pipeline.

Per-trial reconstruction performance was computed by convolving the trained decoder with the trial's normalised EEG to reconstruct the feature, then computing the Pearson correlation (r) between the reconstructed and actual feature. For the mel spectrogram, which has a frequency dimension, the per-trial correlation was averaged across the 8 mel bands to produce a single r value.

Figure 4 shows an example decoding TRF.

3.5 Results Aggregation and Quality Check

For each subject, both encoding (31 feature combinations) and decoding (5 features) models were fit across all frequency bands, and the entire procedure was repeated for both feature pipelines. This gave a total of:

$$20 \text{ subjects} \times 5 \text{ bands} \times 2 \text{ pipelines} \times (31 + 5) \text{ models} = 7,200 \text{ trained TRF models.}$$

After fitting, results were collected into a single CSV file with one row per condition as described above, totalling 7,200 rows. Each row stored the per-trial Pearson correlations, the mean correlation across all trials, the boosting fit time, and metadata (subject

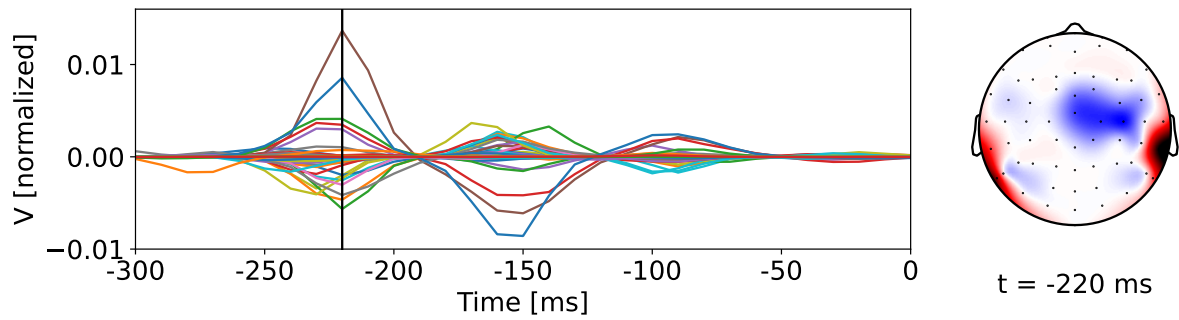


Figure 4: Mean decoding TRF weights across all musicians for processed envelope, theta band. Left: butterfly plot across channels, from -300 to 0 ms. Right: topographic map at peak latency.

ID, group, band, pipeline, modelling direction, feature combination, and number of features in the combination).

Before analysis, the aggregated results were inspected for missing values. Five rows were found to contain only NaN values across all 30 trials. In each case the boosting algorithm had stopped immediately without performing any weight updates, leaving them all at zero. This was verified by checking that the maximum absolute weight in the saved model was 0. The five affected cases were:

- Sub2, alpha band, decoding centroid (both pipelines)
- Sub2, beta band, decoding centroid (both pipelines)
- Sub19, alpha band, encoding pitch alone (raw pipeline only)

These rows were excluded from the analysis.

Chapter 4: Results

4.1 Decoding Results

4.1.1 Per-feature decoding across bands, groups and pipelines

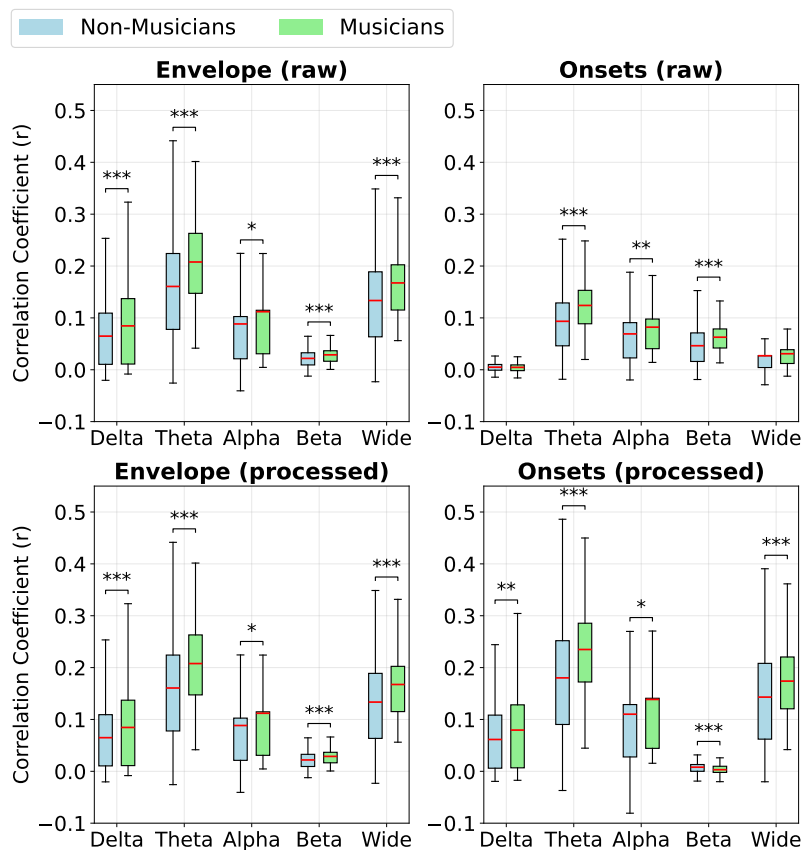


Figure 5: Decoding correlation coefficients across EEG frequency bands for envelope (left) and onsets (right), for both non-musicians (blue) and musicians (green). The top row: raw pipeline. Bottom: processed pipeline. Box plots with red lines showing means. Using t-tests, stars indicate significant differences between the groups (***: $p < 0.001$, **: $p < 0.01$, *: $p < 0.05$). The first four bands of the upper panels were published in Bozilovic & Roman (2025).

Figure 5 shows that the difference between musicians and non-musicians was significant for almost all bands for both envelope and onsets, with musicians reaching higher correlations. The theta band produced the highest values for both features and pipelines. Processing did not improve envelope decoding. For onsets, processing improved performance in all bands except beta, where it decreased.

Figure 6 shows that correlations for pitch, centroid, and mel/MFCC3 are much lower than for envelope and onsets. Significant group differences appear only in some bands, most often in theta, which again gave the highest r values across most features and both pipelines. Processing pitch improved correlations in every band except delta, and made the group difference significant in several bands. For centroid, processing (only standardising) did not improve decoding. MFCC3 showed lower correlations than

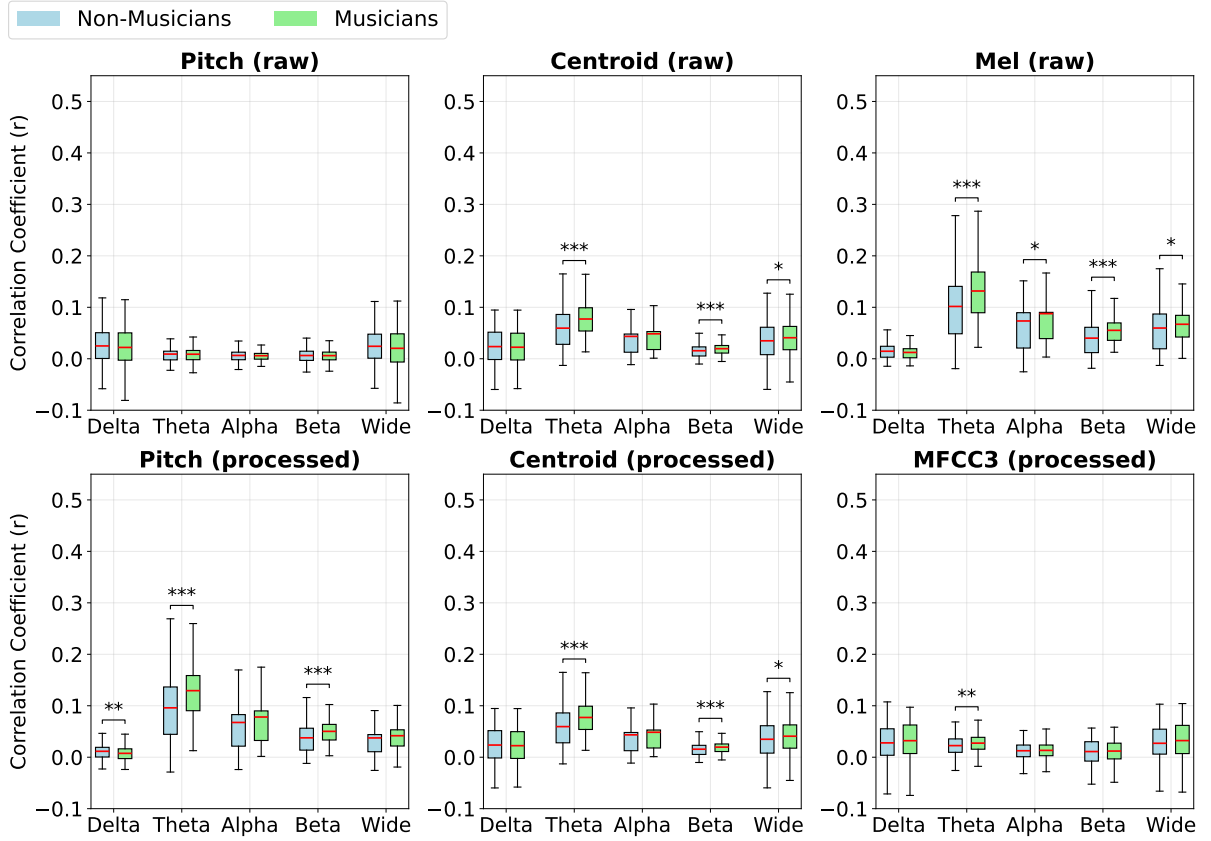


Figure 6: Same plot format as Figure 5 but for pitch (left), centroid (middle), and mel spectrogram/MFCC3 (right), since each appears only in one pipeline

mel, indicating that MFCC3 alone does not capture as much musical information as the full mel spectrogram.

Two-way ANOVAs (group $\times$ band) were run separately for each feature and pipeline to confirm the patterns observed. The group main effect was significant for all features across both pipelines (all $p < 0.003$) except for raw pitch ($p = 0.09$). The band main effect was significant for every feature (all $p < 0.001$, $\eta_p^2 = 0.07 - 0.39$), and significant group $\times$ band interactions appeared for all features except raw pitch and MFCC3. Additionally, two-way ANOVAs (pipeline $\times$ band) were run for each direction (excluding mel and MFCC3 since they are only in one pipeline) and confirmed a significant pipeline effect for decoding ($p < 0.001$, $\eta_p^2 = 0.027$). Full ANOVA tables are available in the supporting material.

4.1.2 Musicians vs non-musicians decoders

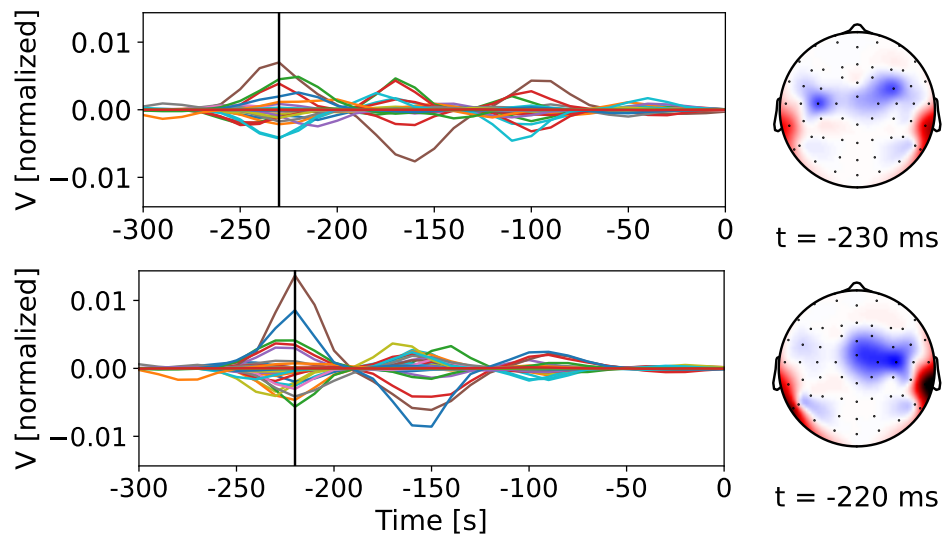


Figure 7: Mean decoding TRF weights for processed envelope in the theta band, averaged within each group (top: non-musicians, bottom: musicians). Left: butterfly plot showing the decoder weights. Right: topographic map at the peak latency. This figure was previously published in Bozilovic & Roman (2025).

Figure 7 shows musicians had larger decoder weights (amplitudes) and stronger corresponding EEG voltage fields (darker colors on the topographic map), particularly over right-lateralised channels. The remaining bands demonstrate similar patterns and are shown in Appendix C.

4.1.3 Example reconstructions

Table 1: Best, median, and worst per-trial decoding correlations for musical piece 2 (trials 2, 12 and 22) in the theta band for each processed feature and mel spectrogram. Each entry shows the subject and trial achieving the corresponding r value.

Feature	Pipeline	Type	Subject	Trial	r
envelope	processed	best	Sub16	2	0.4425
		median	Sub20	22	0.2437
		worst	Sub2	2	-0.0201
onsets	processed	best	Sub16	2	0.5008
		median	Sub8	22	0.2551
		worst	Sub7	22	-0.0077
pitch	processed	best	Sub16	2	0.2656
		median	Sub19	2	0.1466
		worst	Sub7	22	-0.0044
centroid	processed	best	Sub16	2	0.1559
		median	Sub19	2	0.0760
		worst	Sub2	12	-0.0088
mfcc3	processed	best	Sub16	2	0.0619
		median	Sub20	2	0.0250
		worst	Sub2	2	-0.0222
mel	raw	best	Sub16	2	0.2586
		median	Sub15	12	0.1418
		worst	Sub7	22	-0.0078

Table 1 shows that the best reconstructions consistently came from Sub16, a musician, while the worst came from Sub2 and Sub7, both non-musicians. The pattern across features follows the boxplot results (Figures 5 and 6). Envelope and onsets produce the highest r values, pitch and mel spectrogram are intermediate, and centroid and MFCC3 are the lowest.

Figure 8 shows the corresponding reconstructions. The best trials for envelope and onsets follow the contour and timing of the true signal in the last second of the segment. Median trials show partial alignment in the last 0.5 seconds, while worst trials show no alignment. Pitch and mel spectrogram show a similar but much weaker pattern. Centroid and MFCC3 show almost no alignment across all trials, except for a small one around 7.2 seconds in the best and median centroid trials.

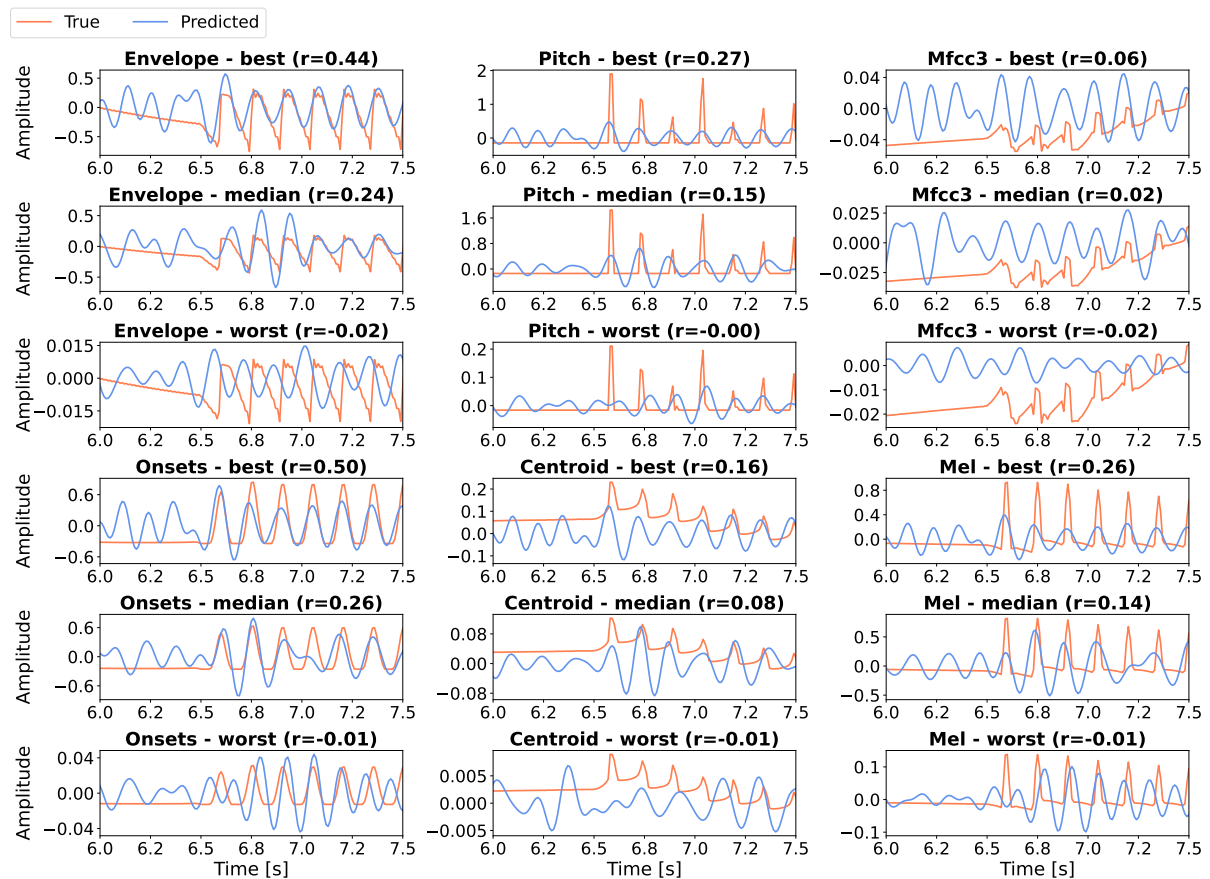


Figure 8: A 1.5 seconds segment of the best, median, and worst per-trial reconstructions for each feature listed in Table 1. Each column shows two features (top to bottom: envelope and onsets, pitch and centroid, MFCC3 and mel spectrogram). The true feature is shown in orange and the reconstructed in blue. For mel, both signals are averaged across the 8 mel bands.

4.2 Encoding Results

4.2.1 Single-feature encoding across bands, groups and pipelines

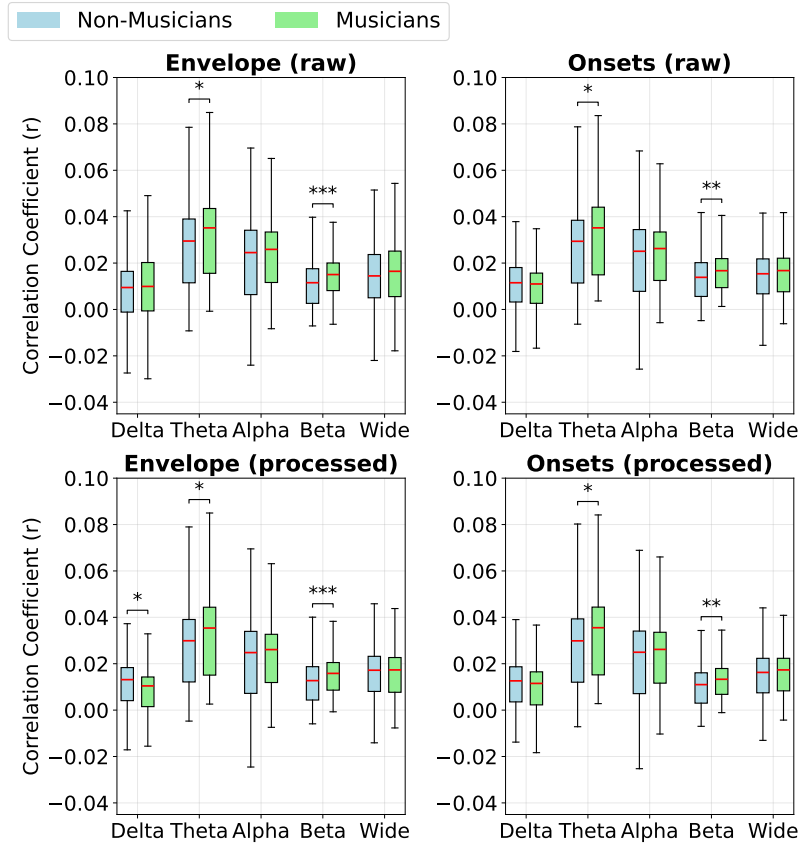


Figure 9: Same plot format as Figure 5 for single-feature encoding correlation coefficients

Figure 9 shows that encoding correlations are much lower than decoding correlations. The difference between musicians and non-musicians is significant for several bands (theta and beta consistently), with musicians reaching higher correlations. The theta band produced the highest values for both features and pipelines. Processing did not improve envelope or onsets encoding.

Figure 10 shows that correlations for pitch, centroid, mel spectrogram, and MFCC3 are again lower than for envelope and onsets. Theta produced the highest values except for raw pitch and mel. Significant group differences appear only in some bands, mostly in theta. Processing pitch made the group difference significant in theta and beta, similar to decoding. Processing centroid (only standardising) did not improve encoding. MFCC3 showed correlations similar to mel, with small improvements in some bands.

Two-way ANOVAs (group $\times$ band) were run separately for each feature and pipeline to confirm the patterns observed. The group main effect was significant for all features (all $p < 0.05$) except raw pitch ($p = 0.94$) and mel ($p = 0.07$). The band main effect was significant for every feature (all $p < 0.001$, $\eta_p^2 = 0.02 - 0.15$). Significant group $\times$ band interactions appeared only for processed envelope and processed pitch. Additionally, two-way ANOVAs (pipeline $\times$ band) were run for each direction (excluding mel and

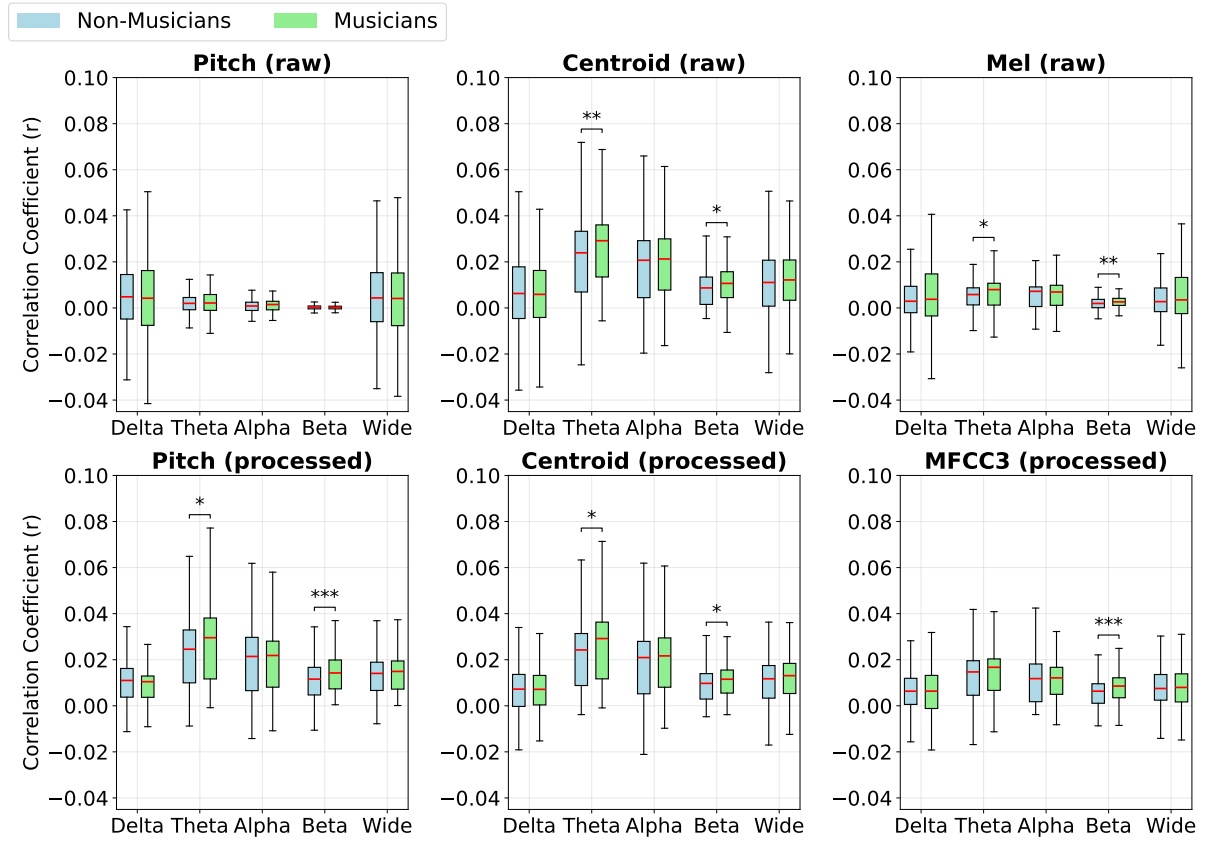


Figure 10: Same plot format as Figure 9 but for pitch (left), centroid (middle), and mel spectrogram/MFCC3 (right), since each appears only in one pipeline.

MFCC3 since they are only in one pipeline) and confirmed a significant pipeline effect for encoding ($p < 0.001$, $\eta_p^2 = 0.011$), a much smaller effect than for decoding.

4.2.2 Effect of combining features

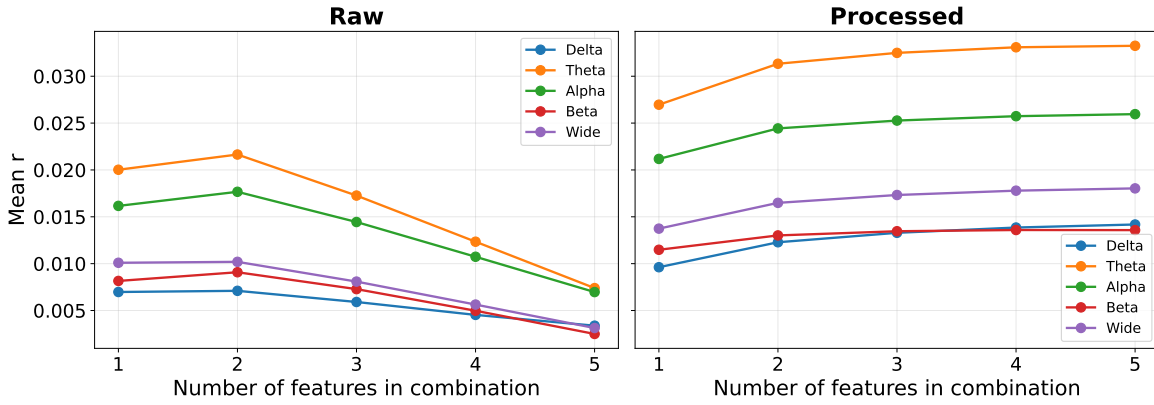


Figure 11: Mean encoding correlation coefficients across EEG frequency bands as the number of features in the model increases from one to five, for the raw pipeline (left) and processed pipeline (right). Each point is the mean r across all feature combinations with the given number of features.

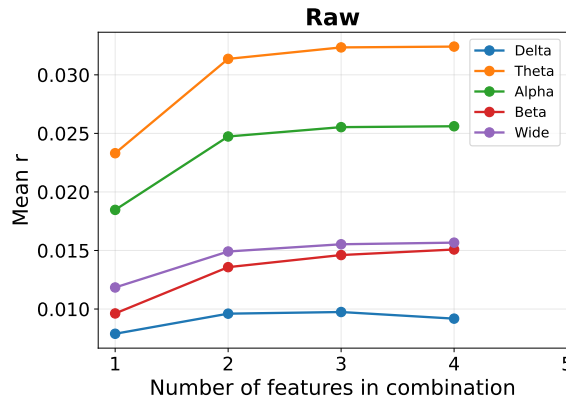


Figure 12: Mean encoding correlation coefficients across EEG frequency bands as the number of features in the model increases from one to four, for the raw pipeline. All combinations containing mel spectrogram were excluded.

Figure 11 shows that adding features decreases encoding correlations in the raw pipeline but increases them in the processed pipeline. Figure 12 shows the same plot for the raw pipeline with all combinations excluding mel spectrogram. Correlations now improve as features are added, matching the processed pipeline. Thus, the decrease in the raw pipeline is driven by mel spectrogram. Across all three plots, the bands follow the same order from highest to lowest r : theta, alpha, wideband, beta, delta. Small deviations appear at four and five features in the processed pipeline (delta exceeds beta) and at five features in the raw pipeline (delta exceeds beta and wideband), but in each case the differences between these bands are small.

One-way ANOVAs were run to test whether the number of features in the model affects encoding correlations, for each band and pipeline. The main effect of number of features was significant for every band and pipeline (all $p < 0.001$, $\eta_p^2 = 0.004 - 0.030$). Post-hoc Bonferroni comparisons showed that in the raw pipeline most pairwise comparisons of number of features were significant, while in the processed pipeline the significant differences were mostly between one feature and the rest.

4.2.3 Feature contributions

To quantify each feature's contribution to encoding performance, a multiple linear regression was fitted separately for each band and pipeline (10 regressions in total). Every individual r value from every feature combination was used as a separate observation, with five binary indicators (0 if feature is absent, 1 if present in the combination) as predictors:

$$r = \beta_0 + \beta_1 \cdot has_env + \beta_2 \cdot has_ons + \beta_3 \cdot has_pit + \beta_4 \cdot has_cen + \beta_5 \cdot has_mel / has_mfcc3 + \epsilon \quad (3)$$

where β_0 is the baseline r , $\beta_1, \dots, \beta_5$ are the coefficients for each feature, the $has_...$ terms are the binary indicators, and ϵ is the residual error.

Each regression finds the set of β 's that minimises the squared error between predicted and actual r 's across all observations. A positive coefficient means including that feature increased r on average, a negative one means it decreased r .

Table 2: Multiple regression coefficients for each feature's contribution to encoding correlation (r), per band and pipeline. Significance: ***: $p < 0.001$, **: $p < 0.01$, *: $p < 0.05$, ns: not significant.

Pipeline	Band	envelope	onsets	pitch	centroid	mel/MFCC3
raw	delta	0.0002 ns	0.0013 ***	-0.0003 ns	-0.0001 ns	-0.0058 ***
	theta	0.0033 ***	0.0030 ***	-0.0015 ***	0.0015 ***	-0.0221 ***
	alpha	0.0022 ***	0.0024 ***	-0.0012 ***	0.0012 ***	-0.0161 ***
	beta	0.0013 ***	0.0025 ***	-0.0007 ***	0.0005 ***	-0.0104 ***
	wide	0.0012 ***	0.0017 ***	-0.0006 **	0.0003 ns	-0.0111 ***
processed	delta	0.0017 ***	0.0020 ***	0.0016 ***	0.0004 ns	0.0004 *
	theta	0.0035 ***	0.0035 ***	0.0011 **	0.0009 *	-0.0006 ns
	alpha	0.0025 ***	0.0029 ***	0.0008 *	0.0007 *	-0.0006 ns
	beta	0.0018 ***	0.0006 ***	0.0006 ***	0.0001 ns	-0.0003 ns
	wide	0.0023 ***	0.0021 ***	0.0011 ***	0.0004 ns	-0.0002 ns

Table 2 shows that envelope and onsets have the highest positive significant effect on encoding correlations (and thus performance), especially in the theta band. The effect of pitch is significant and negative in the raw pipeline but significant and positive in the processed pipeline. Centroid either has no significant contribution or a small positive one. Mel has a strong significant negative effect across all bands (the reason for switching to MFCC3), while MFCC3 mostly has non-significant contribution.

The complete ranking of all 31 encoding feature combinations by mean correlation per band and pipeline is available in the supporting material ('visualisations' notebook).

4.3 Performance vs computational time

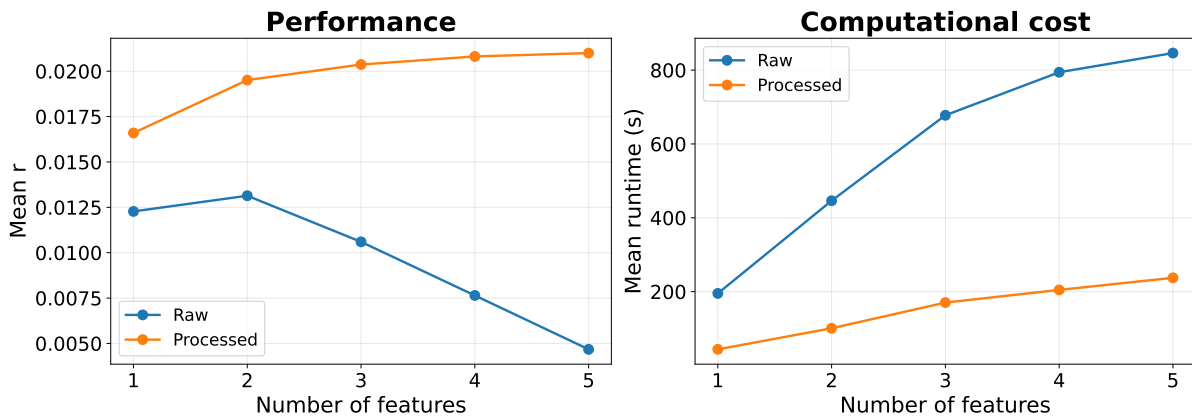


Figure 13: Mean encoding correlation r (left) and mean boosting fit time in seconds (right) per number of features in the encoding model, for the raw pipeline (blue) and the processed pipeline (orange). Each point is averaged across all subjects, bands, and feature combinations with the given number of features.

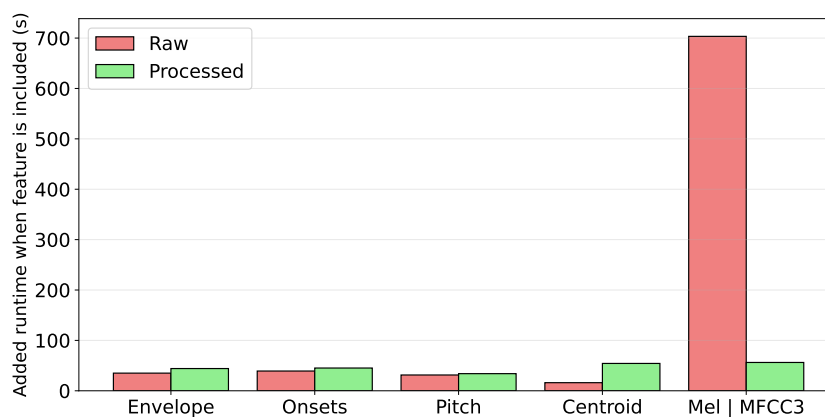


Figure 14: Average runtime (in seconds) added per feature in encoding models, for raw (pink) and processed (green) pipeline. Each bar: the difference between mean runtime of combinations including the feature and excluding it. Mel spectrogram and MFCC3 share the fifth bar (each only in one pipeline).

Figure 13 shows that adding features increases boosting fit time, especially in the raw pipeline. The figure also shows averaged performance, which was previously displayed per band (Figure 11) but is shown here as a single average to make it easily comparable with the computational cost. In the processed pipeline performance and runtime both increase with more features, while in the raw pipeline they move in opposite directions because of mel spectrogram, as established earlier.

Figure 14 shows that mel spectrogram in the raw pipeline is by far the most expensive feature to include. The other four features in the raw pipeline have similar, much smaller contributions (centroid being the smallest). All features in the processed pipeline are similarly balanced and have larger runtime contributions than the same features in the raw pipeline (except MFCC3).

As Table 3 shows, the same pattern is visible in both directions of TRF modelling.

Table 3: Mean boosting fit time per model (in seconds) for both directions and both pipelines. The 'without mel' rows show the raw pipeline average after excluding all combinations containing the mel spectrogram.

Direction	Pipeline	Set	Mean (s)
Encoding	raw	all	549.5
	raw	without mel	186.4
	processed	all	135.2
Decoding	raw	all	2926.9
	raw	without mel	2084.9
	processed	all	906.8

The raw pipeline was more expensive than the processed pipeline in both directions. Excluding mel spectrogram brought raw pipeline's runtimes closer to the processed pipeline's, especially for encoding. Decoding was considerably slower than encoding overall.

Chapter 5: Evaluation

5.1 Methodological discussion

5.1.1 Strengths

The dataset set used is openly available, provides balanced data between musicians and non-musicians, and offers enough data per subject for TRF models (Di Liberto et al. 2020).

Six acoustic features were analysed. Existing studies on this dataset focuses only on envelope, onsets, and melodic expectations (Di Liberto et al. 2020, Galeano-Otálvaro et al. 2024, Kern et al. 2022). Other audio–EEG TRF studies similarly concentrate on temporal features, most commonly the envelope alone (Lalor & Foxe 2010, Accou et al. 2023, Drennan & Lalor 2019, Di Liberto et al. 2021), sometimes combined with melodic expectations (Di Liberto et al. 2020, Kern et al. 2022), with spectral and cepstral features rarely included (Di Liberto et al. 2015). Including pitch, spectral centroid, mel spectrogram, and MFCC3 made it possible to test how spectral and cepstral features contribute to TRF performance, which has not previously been assessed on this data.

The feature extraction pipeline itself was treated as an experimental variable. This isolated the effect of the processing steps which a single pipeline design cannot do.

Both encoding and decoding were performed across five frequency bands. Not restricting to one band allowed band-specific effects to be isolated. Running both modelling directions made it possible to assess both how well neural responses can be predicted and how well the stimulus can be reconstructed using the same data.

The encoding analysis covered all 31 non-empty feature combinations per subject, band, and pipeline. This allowed each feature's individual contribution to be quantified through a multiple linear regression.

Several stages of the pipeline (EEG preprocessing, stimulus–EEG segmentation, and per-trial normalisation) follow the reference TRF procedure in Eelbrain's documentation (Brodbeck et al. 2023), which reduces the risk of implementation errors (for example hyperparameter tuning) that can affect TRF analysis (Crosse et al. 2021).

Finally, since the boosting algorithm used for TRF estimation tends to leave redundant weights at zero (Brodbeck et al. 2023), it makes the multi-feature encoding analysis more interpretable than ridge regression would have been.

5.1.2 Limitations and Improvements

The dataset (Di Liberto et al. 2020) does not include the audio files that participants listened to, only the MIDI files of the stimuli. Audio for feature extraction was therefore rendered from MIDI using built-in synthesiser, which matches the way the original stimuli were created (Di Liberto et al. 2020) but does not reproduce them exactly. Synthesised piano sounds has a simplified timbre compared to a recorded acoustic instrument. Features such as the spectral centroid, mel spectrogram, and MFCC3 therefore have less informative information, which could explain part of their weaker contribution

to TRF performance. A dataset that published the exact stimulus audio (for example WAV files) would allow these features to be tested on a more representative signal.

The stimuli are monophonic Bach pieces played with a single piano timbre. This restricts the generalisability of the findings, since music in real listening conditions typically involves multiple instruments, polyphony, and a wider range of timbres and genres. To generalise these findings, the pipeline should be tested on additional music-EEG datasets with more naturalistic musical stimuli.

The main limitation of TRF is that linear models assume a linear stimulus-response relationship and cannot capture the non-linear processing that happens in the auditory pathway (Wang et al. 2008, Puffay et al. 2023). The modest correlations in results leave space for non-linear methods to extract additional signal, particularly for spectral and cepstral features where TRF performance was the weakest.

The same TRF hyperparameters were used across all bands, features, pipelines, and subjects. Optimal hyperparameter values can vary with feature type and frequency (Crosse et al. 2021), so a per-condition hyperparameters might have improved some configurations.

5.2 Results discussion

5.2.1 Strengths

The results show several patterns that are robust across many independent comparisons. Musicians produce significantly higher correlations than non-musicians for almost all bands and features in decoding (except raw pitch and MFCC3) and several bands and features in encoding. The theta band is usually the best-performing band across both groups, pipelines and directions. Since these effects are replicated across features, bands, pipelines, and modelling directions rather than emerging from one test, they are unlikely to be artefacts of a particular analysis choice.

The two pipeline design separated two kinds of feature processing. Steps that change the shape of a feature (onsets and pitch) and steps that only standardise its scale (envelope and centroid). The shape-changing steps improved decoding for the features in most bands, and for pitch the improvement is seen in encoding as well. The standardisation-only steps did not improve performance. This clarifies that the improvements from the processed pipeline come from changing the shape of features, not from standardisation. This could not be observed without comparing two pipelines.

For encoding, the feature-combination analysis showed that adding features improves performance in the processed pipeline, and in the raw pipeline once the mel spectrogram is excluded. The regression identifies the mel spectrogram is the feature that drags multi-feature encoding performance down, and the processed pitch is the feature whose processing actually changes the direction of its contribution. These claims about individual feature contributions are the ones that the boxplots and ANOVAs on single-feature results alone could not make.

The decoder topographies revealed a spatial difference between groups that the per-feature correlations alone could not show. Musicians had stronger right-lateralised voltage fields, and the same right-lateralised pattern is visible across all five bands

(Figure 7 and Appendix C). The fact that the lateralisation generalises across bands, rather than only appearing in the best-performing one, suggests that this is a consistent finding of how musicians' brains track music, not an effect of theta-band performance. This observation is consistent with prior findings of hemispheric lateralization observed in musicians Koelsch et al. (2002).

The combined performance and runtime analysis (Figures 13 and 14, Table 3) shows that the processed pipeline is both faster and produces stronger results. The mel spectrogram does not only contribute negatively to results, but it makes the raw pipeline computationally more expensive. This means that future TRF analyses on similar data should use the processed pipeline, and should avoid the mel spectrogram unless its specific properties are needed for the research question.

Finally, the statistical analysis used multiple complementary procedures rather than relying on a single test. Pearson's r between predicted and actual signals was used as the primary metric, in line with standard TRF studies (Crosse et al. 2016, Brodbeck et al. 2023). Independent-samples t -tests assessed musician versus non-musician differences for each individual condition, two-way ANOVAs tested group $\times$ band and pipeline $\times$ band effects, multiple linear regressions quantified each feature's contribution to encoding, one-way ANOVAs tested whether the number of features matters and post-hoc Bonferroni comparisons followed up the significant effects. The results of this thesis emerge from all of these analyses, not from one. The consistency of results across independent statistical approaches gives the conclusions a stronger basis than any single test would.

5.2.2 Limitations and Improvements

The correlation values are modest, especially for encoding, where even the best correlations are below 0.1. Encoders predict each of the 64 EEG channels separately from a stimulus feature, whereas decoders combine all 64 channels to reconstruct a stimulus (lower dimension, less noisy), which is partly why encoding correlations are systematically lower. The lower values mean the predictions are useful for comparing conditions, but cannot be interpreted as accurate signal reconstruction. Smaller group, band, and pipeline differences are harder to detect on the encoding side. Several encoding ANOVAs (pipeline and pipeline $\times$ band interactions) produced small effect sizes (η_p^2) despite significant p -values. Thus, the practical importance of the pipeline effect for encoding is limited compared to the band and group effects. Future work could report encoding performance at the most predictive channels (for example, the top auditory-responsive electrodes), which could potentially raise the correlation values closer to those typically reported in the literature (Crosse et al. 2021).

The mel spectrogram produced poor encoding results, with a strongly negative regression coefficient across all bands. This may be partly an artefact of how the multi-band mel spectrogram interacts with boosting on a limited amount of training data. The algorithm may have stopped early before fitting useful weights, leading to predictions worse than those of other features. Using fewer mel bands, using more training data per subject, or both, could help the boosting algorithm fit useful weights for higher-dimensional features like mel spectrogram (Mesik & Wojtczak 2023).

Five TRF models stopped immediately during boosting without performing any weight

updates and were excluded from the analysis. Although the exclusion only affected five rows out of 7,200 models, this indicates that boosting is not fully robust for every subject, feature and pipeline configuration. Re-fitting the affected models with ridge regression (Crosse et al. 2016), or with per-condition boosting hyperparameters, could reveal whether the issue can be fixed.

5.2.3 Relation to previous studies

The results align with several findings in the literature. The low-frequency bands consistently produced the highest TRF performance for envelope and onsets. This matches the role of theta activity in tracking the rate of individual notes in music (Doelling & Poeppel 2015), and the 1–8 Hz range is consistently the most informative in audio–EEG studies (Di Liberto et al. 2020, Galeano-Otálvaro et al. 2024). Splitting this range into delta and theta separately showed that within it theta carries the most stimulus information, especially for envelope and onsets, which the combined-band approach used in earlier work could not reveal.

The musician advantage in decoding and encoding matches what previous work on this dataset has shown for encoding. Di Liberto et al. (2020) reported larger melodic-expectation responses in musicians at frontal EEG channels, and Galeano-Otálvaro et al. (2024) found that melodic expectations enhanced EEG reconstruction more strongly in the beta band for musicians. This thesis extends those findings to more acoustic features (pitch, centroid, mel, MFCC3) and to the decoding direction, neither of which has been examined on this dataset before.

Decoding correlations were much higher than encoding correlations, consistent with the general property of the two modelling directions reported by Crosse et al. (2021). The decoding correlations reported in this thesis are in line with what has been reported for linear envelope decoding on speech–EEG data, where the median Pearson correlation is around 0.19 (Accou et al. 2023). For envelope and onsets, the decoding correlations in this thesis are around 0.2, comparable to this median. For other features (pitch, centroid, mel/MFCC3), correlations are lower, around 0.05.

The encoding correlations are modest, all below 0.1 (usually around 0.03 for envelope and onsets and 0.02 for other features). This is in the lower range of those reported for the same dataset by Di Liberto et al. (2020), Galeano-Otálvaro et al. (2024). Higher prediction accuracies have been reported using invasive recording techniques such as ECoG (electrocorticography), where correlations on the same musical stimuli reach around $r = 0.3$ in temporal cortex (Di Liberto et al. 2020).

5.3 Legal, social, ethical, and sustainability considerations

5.3.1 Legal

This thesis uses a publicly available dataset (Di Liberto et al. 2020). No new data collection was performed, so no human participants were recruited. The original study obtained written informed consent from all 20 participants and was approved by the CERES committee of Paris Descartes University (Di Liberto et al. 2020). The dataset is anonymised, contains no identifying information, and was made available for reuse by the original authors, so no separate ethical approval was required for this project.

All software used (Python, NumPy, SciPy, PrettyMIDI, Soundfile, Librosa, MNE-Python, Eelbrain, Pandas, Statsmodels, Pingouin, and Matplotlib) is open source and freely available for use. AWS was used for computational resources, in accordance with its standard terms of service.

All work not produced by the author is appropriately cited in the references, in accordance with QMUL's academic integrity policy.

5.3.2 Social

The results presented in this thesis show that musicians' brain responses track acoustic features of music more strongly than those of non-musicians. There is a risk this is read as implying that non-musicians do not understand or enjoy music in a meaningful way, which would be a misreading. The analysis measures a quantitative difference in how reliably acoustic features can be recovered from the EEG signal using a linear model. It does not measure musical enjoyment. Non-musicians in this thesis still produced decodable signals, but the TRF reconstructed them less accurately. Communication of these results should avoid suggesting that stronger neural tracking means a more 'real' experience of music.

5.3.3 Ethical

The sample used in the dataset is small: 20 participants in total, 10 per group. With this sample size, any effect of musical expertise estimated here carries uncertainty, and the findings should not be extrapolated to all musicians or all non-musicians. The participants were also aged between 23 and 42 (Di Liberto et al. 2020), so the conclusions may not generalise to children, adolescents or older adults. Additionally, the participants are expert pianists trained in Western classical music, and the stimuli are from the same genre (Di Liberto et al. 2020). The conclusions are therefore limited to Western classical music, and extending them to other musical genres would not be appropriate.

5.3.4 Sustainability

The TRF pipeline is computationally expensive, particularly for the raw pipeline. Running the full set of 7,200 models required AWS compute instances, which have a non-zero carbon cost. However, the pipeline can also be executed on a standard local machine (with substantially longer runtimes), so the AWS use is not a requirement of the methodology. As discussed in Chapter 5, a non-linear modelling approach would have required substantially more compute, so the choice of linear TRFs also has a sustainability benefit.

The computational analysis included in this thesis (Figures 13 and 14, and Table 3) makes the cost of each feature explicit, which can help future work avoid expensive feature combinations that do not improve results.

All code and results are released on GitHub alongside the thesis (<https://github.com/ZorkaBozilovic/music-eeg-modelling>), so that other researchers can re-run, extend, or reuse the pipeline without having to repeat all the computational work.

Chapter 6: Conclusion

6.1 Answers to the research questions

RQ1: Can the melody someone is listening to be reconstructed from their neural data? To a meaningful extent for temporal features, with Pearson correlations around 0.2. Spectral and cepstral features were reconstructed less reliably, with correlations around 0.05.

RQ2: Can neural data be predicted from the music someone is listening to? Not to a meaningful extent. Encoding correlations were around 0.03 for envelope and onsets, around 0.02 for all other features. The predictions are useful for comparing conditions but cannot be interpreted as accurate signal reconstruction.

RQ3: Does increasing the number of acoustic features in encoding models improve prediction performance? Yes, in the processed pipeline, and in the raw pipeline once the mel spectrogram is excluded. The multiple regression identified envelope and onsets as the strongest positive contributors.

RQ4: How does encoding and decoding performance vary across frequency bands of the neural data? Yes. The theta band gave the highest correlations across most features, both groups, both pipelines, and both modelling directions. Across encoding feature combinations, the bands followed the same order from highest to lowest correlation: theta, alpha, wideband, beta, delta.

RQ5: Does musical expertise lead to measurable differences in the neural responses to music? Yes. Musicians produced significantly higher correlations than non-musicians across nearly all bands and features in decoding, and several bands and features in encoding. The decoder topographies also showed stronger right-lateralised voltage fields in musicians across all five bands.

The corresponding hypotheses set out in Chapter 1 were supported, except the second one ('Neural data can be predicted from music to a meaningful extent'), which was not.

6.2 Achievement of Objectives

All objectives were completed:

- Relevant literature on neural responses to music and TRF modelling was reviewed in Chapter 2, identifying the gaps this thesis addresses.
- The data was prepared as described in Chapter 3: audio was rendered from MIDI, six acoustic features were extracted in two pipelines, and the EEG was downsampled and split into five frequency bands.
- Forward and backward TRF pipelines were implemented, with encoding TRFs fitted for all 31 non-empty feature combinations per pipeline and decoding TRFs fitted for each feature individually, producing 7,200 trained models in total.
- Single-feature decoding and single and multi-feature encoding were evaluated across all five frequency bands, with feature contributions quantified through a

multiple linear regression.

- TRF performance was compared between musicians and non-musicians for every feature, band, and pipeline, with the comparison extended to decoder topographies.

6.3 Achievement of Aim

The aim was achieved. The effect of different acoustic features was investigated through decoding all six features individually and encoding all 31 non-empty feature combinations per pipeline, with envelope and onsets producing the highest correlations. The best-performing frequency band was identified as theta, which gave the highest correlations across most features, both pipelines, and both modelling directions. Musicians produced stronger neural responses and more right-lateralised decoder topographies than non-musicians across all five frequency bands. In doing so, this thesis addressed gaps in previous music–EEG TRF research, where decoding had not been performed on this dataset, feature combinations had not been systematically examined, and performance across individual frequency bands had not been compared across this range of acoustic features (Di Liberto et al. 2020, Galeano-Otálvaro et al. 2024).

6.4 Future work

Several extensions would build on this work. Non-linear models such as artificial neural networks could be compared against the linear TRFs used here on the same dataset, particularly for spectral and cepstral features where linear performance was the weakest. Additionally, this pipeline and artificial neural networks can be tested on another dataset with more data. Higher-level musical features, such as the surprise and entropy derived from IDyOM could be added to the framework to capture both acoustic and higher-level feature and their combinations. Also, extending the analysis to polyphonic music and non-Western classical music would test whether the patterns observed here are generalisable.

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Appendix A: Glossary

ANOVA Analysis of variance. A statistical test used to compare group means across conditions.

Bandpass filter A filter that keeps a chosen range of frequencies and removes the rest.

Bonferroni post-hoc A correction applied to multiple pairwise comparisons after an ANOVA.

Butterfly plot A plot showing all EEG (or TRF) channels overlaid on the same axes.

Centroid Spectral centroid.

CNN Convolutional neural network.

CPU Central processing unit.

CSV Comma-Separated Values.

E3–C7 The pitch range used to bound F0 estimation. Spans from E in the third octave to C in the seventh octave.

GPU Graphics processing unit.

Hilbert transform A mathematical operation. Used here to extract the amplitude envelope of a signal.

Interpolation Filling in missing values in a signal by estimating them from neighbouring values. Used here for pitch frames where pYIN returns NaN.

Mel Mel spectrogram.

MIDI Musical Instrument Digital Interface. A symbolic format that encodes notes rather than audio waveforms.

NaN Not a Number. A placeholder value indicating that no valid number is available.

Pearson correlation A measure of linear dependence between two signals, ranging from -1 to $+1$.

RAM Random-access memory.

t-test A statistical test comparing the means of two groups. Used here for musicians vs non-musicians.

Time lag The delay between the stimulus and the corresponding neural response, represented as a separate weight in the TRF.

Topographic map A scalp plot showing the spatial distribution of EEG (or TRF) values across electrodes.

vs Versus.

Appendix B: Learning Outcomes of Programme of Study

This project integrates key outcomes of the BSc Computer Science and Mathematics programme by combining mathematical modelling, data analysis, and software development. It applies machine learning and signal processing techniques to real EEG data, strengthening skills in algorithmic design, statistical reasoning, and computational implementation.

In terms of academic content, it applies the mathematical, scientific, and engineering elements of computer science to a problem in auditory neuroscience, an application area at the intersection of computing and the sciences. The work also reflects on the broader professional context of computer science through its legal, social, ethical, and sustainability discussion.

In terms of disciplinary skills, the project required problem-solving across feature extraction, EEG preprocessing, TRF modelling, statistical analysis of 7,200 model outputs, and recognising methodological risks.

In terms of attributes, the project was managed independently over the academic year, including writing the thesis and documenting and releasing the code repository for reuse.

Beyond these programme-specific outcomes, the project also strengthens transferable skills emphasised by the programme: critical evaluation of competing methods, and scientific communication, including the publication of part of this work at the AES International Conference on Artificial Intelligence and Machine Learning for Audio.

Appendix C: Decoding TRFs Across Bands

The following figures show mean decoding TRF weights for processed envelope across the four EEG bands not shown in the main text (delta, alpha, beta, and wideband). Each figure follows the same format as Figure 7: top row shows non-musicians, bottom row shows musicians, the left panels show butterfly plots across channels, and the right panels show topographic maps at the peak latency.

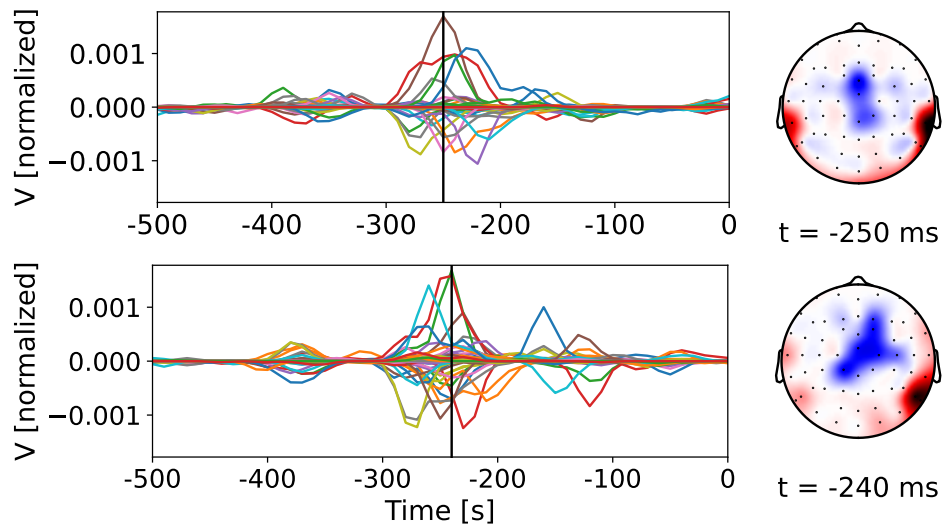


Figure 15: Mean decoding TRF weights for processed envelope in the delta band.

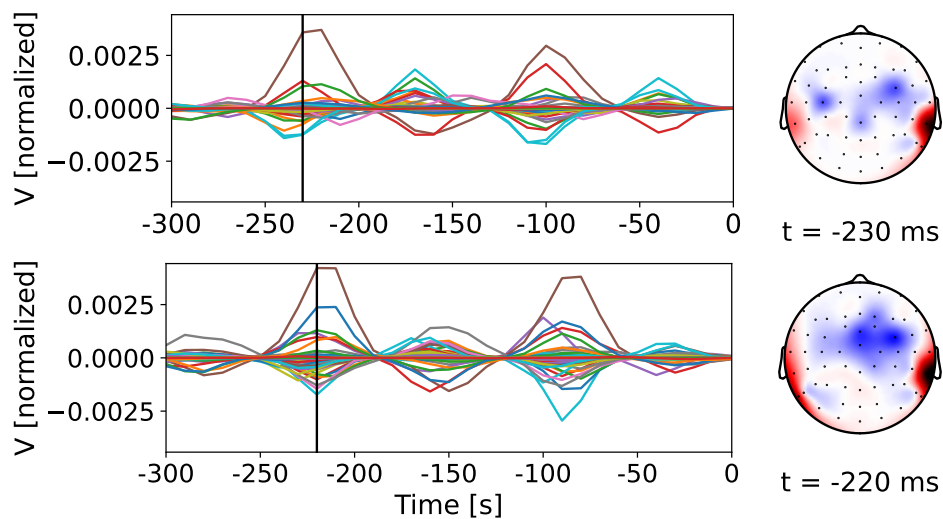


Figure 16: Mean decoding TRF weights for processed envelope in the alpha band.

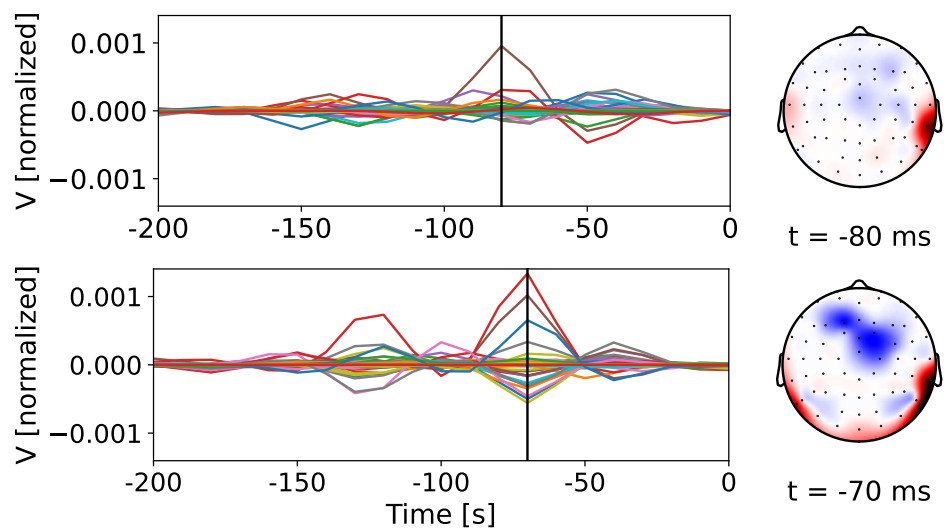


Figure 17: Mean decoding TRF weights for processed envelope in the beta band.

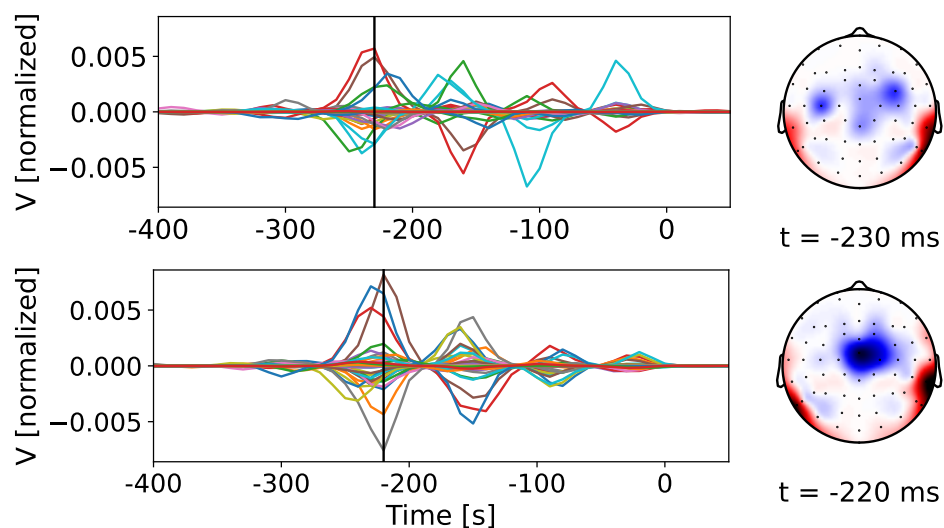


Figure 18: Mean decoding TRF weights for processed envelope in the wideband.

Appendix D: Word Count

The total word count for the main body of this report is 10,973 words. This includes the abstract and the main body text (Chapters 1–6), and excludes the tables of contents, references, and appendices, as required by the 'UG Project Handbook'.

Appendix E: Generative AI Tools Statement

ECS635U – Project Module

Generative AI Tools – Student Accountability Statement

The completed and signed statement must be included in the appendix of the progress and final reports.

Student Name: Zorka Bozilovic

Project Title: Linear Encoding and Decoding of Brain Activity in Response to Music

Supervisor Name: Dr. Iran R Roman, PhD

1. Have you used Generative AI tools for your project?

YES

If yes, please complete sections 2 and 3 and sign the declaration below.

If no, please sign the declaration below.

2. Please provide details of all the AI tools that you used in your project work.

AI Tool	Version	Publisher	Access URL
Claude	Sonnet 4.6	Anthropic	https://claude.ai

3. If you have used Artificial Intelligence (AI) tools in any part of your project please include in table format how, where, and why these tools were used. This should include references to the relevant parts of your project work — for example:

- Specific chapters or sections of the report (e.g. literature review, methodology, requirements)
- Software development (e.g. AI-assisted code generation, debugging, documentation)
- Interview or survey materials (e.g. AI-generated or refined questions/prompts)
- Datasets (e.g. AI-assisted data cleaning, labelling, or synthesis)
- The showcase video (e.g. AI-assisted scripting, narration, or editing support)

Be specific and transparent about your use of AI tools, including both direct outputs (e.g. text, code, images) and indirect uses (e.g. idea generation, editing assistance). The goal is to ensure academic integrity and provide a clear record of how AI contributed to your work.

Example Table – please add your own

Project Section	Description	Justification
Familiarising with a new topic	Used to get a broad understanding of neuroscience topics	As allowed below, I used GenAI tools for familiarising myself with new topics before reading neuroscience papers
Report and Code Comments	Used to improve writing structure	As allowed below, I used GenAI tools for checking grammar and spelling, and for improving writing structure.

3. Generative AI Student Declaration

- I have declared all uses of AI tools (e.g., coding assistants, text generators, data analysis tools) in this project.
- I have reviewed and verified all AI-assisted content to ensure accuracy and suitability.
- I take full responsibility for the originality and integrity of my project submission.
- I have referenced all sources, including those suggested or produced with AI support.
- I have not used AI in ways that undermine academic honesty (e.g., plagiarism, unverified code submission).
- My use of AI was supplementary and did not replace my own independent learning and problem-solving.

Signed: Zorka Bozilovic Date: 18.05.2026.

Guidance notes for students

- Students are strongly encouraged to review the following resources
 - Student Guide to Gen AI <https://www.qmul.ac.uk/library/academic-skills/student-guide-to-generative-ai/>
 - Academic Integrity and Misconduct Policy <https://www.qmul.ac.uk/governance-and-legal-services/media/dgls-media/policy/current-policies/Academic-Integrity-&-Misconduct-Policy.pdf>
- The use of essay mills, homework help sites, plagiarism, collusion, falsification, impersonation, or actions that may give an unfair advantage is strictly prohibited in your final year project. Please refer to the QMUL Academic Integrity and Misconduct policy above.
- It is never acceptable to use GenAI tools to write your entire literature review and submit it as your own work; doing so is considered a form of plagiarism since GenAI outputs are derived from ideas of human authors without proper referencing.
- You may use GenAI tools for checking grammar and spelling, provided the tool is not rewriting a large portion of your work for your final year project.
- Using GenAI for idea generation or initial planning is generally appropriate but always consider the context and assessment requirements for your project. If in doubt, please talk to your supervisor.
- If you are ever unsure whether GenAI is permitted for any part of your project always ask your project supervisor for explicit clarification.
- Unless stated otherwise, GenAI should only be used to assist your learning at the initial stages of your project and your final submission of your project, and the code must still be substantially your own original work.
- Inappropriate use of GenAI will be viewed as plagiarism under the QMUL academic integrity and misconduct policy this includes submitting work where GenAI's contribution exceeds what is reasonably allowed in the final year project.
- Attempting to pass off GenAI-generated content as your own can result in disciplinary investigation under academic misconduct regulations; you may be asked to attend a viva to confirm authorship.
- Before using GenAI, make sure:
 - You are permitted (or required) to use it for your specific task in your final year project.
 - You fully understand the potential risks, limitations, and how GenAI operates.
 - Your work remains genuinely your own and reflects your understanding.

- Please note that GenAI is helpful for gathering background information, but outputs may be biased, inaccurate, outdated, or fabricated.
- Always consult and cite original, credible sources; GenAI should never be treated as an original source and should not be cited as an author or in the reference list.
- Critically assess all GenAI output for accuracy, double-check any facts, quotations, or citations it suggests especially if you use them in your literature review or other chapters.
- If GenAI has been used for drafting, planning, or structuring, you must clearly acknowledge and document its use in your submission. Therefore, you should keep drafts and records of your own process to show your independent scholarly work and reading notes
- GenAI use may not require acknowledgement for:
 - Familiarising yourself with a topic or research landscape.
 - Summarising information to prepare for more in-depth research.
 - Improving grammar or writing structure.
- Please note that output from GenAI is typically treated as having no author so you cannot cite Gen AI as an author in your report
- When in doubt, transparency and honesty about your methods and sources including GenAI are expected and protect your academic integrity

Appendix F: Acknowledgements

I would like to thank my supervisor, Dr. Iran R. Roman, for his guidance and support throughout this project, and for believing in me from the very beginning, including helping me publish my first paper as part of this work.

I am also grateful to the professors and faculty at Queen Mary University of London for everything they have taught me over the course of my degree.

To my family, thank you for your endless love and support. Thank you to my parents *Ana* and *Zvonko*, to my brothers *Vasilije* and *Djordje*, to *Aleksa*, and to *Baba*, *Deda*, *Miloš*, *Cica*, *Teča*, *Lazar*, *Snajka*, *Lana*, *Luka*, *Teksi*, *Aki*, and *Niki*. Finally, thank you to my friends for always being there. I am so lucky to have all of you.